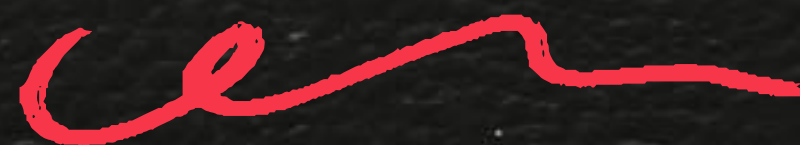




Catching The "137" Serial Killer

Martijn Dashorst

Start →



SD	Productie	Acceptatie	Test	Cursus	Inkijk	Project	filter		16:19:04																																																					
002						2025.12.1	003						2025.12.1	004						2025.12.1	006						2025.12.1	007						2025.12.1	008						2025.12.1																					
SVO							Gilde Opleidingen							Rivor							Vonk (Clusius College)							Vonk (ROC Kop van Noord-Holland)							ROC Midden Nederland																											
Piek sessies: 228 Live sessies: 48							Piek sessies: 431 Live sessies: 247							Piek sessies: 103 Live sessies: 62							Piek sessies: 92 Live sessies: 27							Piek sessies: 118 Live sessies: 66							Piek sessies: 721 Live sessies: 387																											
Duur: 0.039s Load average: 0.1							Duur: 0.121s Load average: 0.4							Duur: 0.090s Load average: 0.2							Duur: 0.036s Load average: 0.1							Duur: 0.041s Load average: 0.3							Duur: 0.159s Load average: 0.5																											
Jobs Running: 0 Jobs Queued: 0							Jobs Running: -1 Jobs Queued: 1							Jobs Running: 0 Jobs Queued: 0							Jobs Running: 0 Jobs Queued: 0							Jobs Running: 0 Jobs Queued: 0							Jobs Running: 1 Jobs Queued: 0																											
Geen jobs							Geen jobs							Geen jobs							Geen jobs							Geen jobs							⚠ BusinessObjects4 Rapportage 35%																											
009						2025.12.1	010						2025.12.1	011						2025.12.1	012						2025.12.1	013						2025.12.1	014						2025.12.1																					
MBO Amersfoort							Alfa College							Firda							ROC Nijmegen							Menso Altling							VISTA college																											
Piek sessies: 235 Live sessies: 168							Piek sessies: 465 Live sessies: 222							Piek sessies: 521 Live sessies: 344							Piek sessies: 229 Live sessies: 150							Piek sessies: 36 Live sessies: 18							Piek sessies: 381 Live sessies: 198																											
Duur: 0.124s Load average: 0.3							Duur: 0.075s Load average: 0.4							Duur: 0.307s Load average: 1.6							Duur: 0.085s Load average: 0.2							Duur: 0.034s Load average: 0.0							Duur: 0.229s Load average: 0.4																											
Jobs Running: 0 Jobs Queued: 0							Jobs Running: 1 Jobs Queued: 2							Jobs Running: 0 Jobs Queued: 0							Jobs Running: 0 Jobs Queued: 0							Jobs Running: 0 Jobs Queued: 0							Jobs Running: 0 Jobs Queued: 0																											
Geen jobs							⚠ Bron bericht versturen 53%							⚠ Rooster importeren 16%							Geen jobs							Geen jobs							Geen jobs																											
							⚠ Bron Bericht processor Job 0%																																																							
015						2025.12.1	017							2025.12.1	020						2025.12.1	021						2025.12.1	022						2025.12.1	026						2025.12.1																				
Yonder							Albeda							Aeres							Scalda							Lentiz							Cibap																											
Piek sessies: 352 Live sessies: 185							Piek sessies: 702 Live sessies: 445							Piek sessies: 153 Live sessies: 84							Piek sessies: 436 Live sessies: 193							Piek sessies: 47 Live sessies: 21							Piek sessies: 63 Live sessies: 38																											
Duur: 0.102s Load average: 1.4							Duur: 0.085s Load average: 0.6							Duur: 0.057s Load average: 0.2							Duur: 0.123s Load average: 0.4							Duur: 0.031s Load average: 0.1							Duur: 0.090s Load average: 0.1																											
Jobs Running: 1 Jobs Queued: 0							Jobs Running: 0 Jobs Queued: 0							Jobs Running: 0 Jobs Queued: 0							Jobs Running: 0 Jobs Queued: 0							Jobs Running: 0 Jobs Queued: 0							Jobs Running: 0 Jobs Queued: 0																											
⚠ Workflowautomaat 30%							Geen jobs							Geen jobs							Geen jobs							Geen jobs							Geen jobs																											
⚠ Documenten Afdrukken bij toetsmomenten job 0%																																																														
033						2025.12.1	035						2025.12.1	036						2025.12.1	040						2025.12.1	041						2025.12.1	042						2025.12.1																					
Horizon							KW1C							Landstede							Ter AA							Amsterdam							Regio College																											
Piek sessies: 463 Live sessies: 303							Piek sessies: 298 Live sessies: 162							Piek sessies: 254 Live sessies: 122							Piek sessies: 149 Live sessies: 90							Piek sessies: 2093 Live sessies: 1348							Piek sessies: 203 Live sessies: 118																											
Duur: 0.183s Load average: 1.1							Duur: 0.123s Load average: 0.1							Duur: 0.059s Load average: 0.2							Duur: 0.056s Load average: 0.1							Duur: 0.138s Load average: 2.9							Duur: 0.081s Load average: 0.2																											
Jobs Running: 0 Jobs Queued: 0							Jobs Running: 0 Jobs Queued: 0							Jobs Running: 0 Jobs Queued: 0							Jobs Running: 0 Jobs Queued: 0							Jobs Running: 0 Jobs Queued: 0							Jobs Running: 0 Jobs Queued: 0																											
Geen jobs							Geen jobs							Geen jobs							Geen jobs							⚠ Rooster importeren 76%							Geen jobs																											
043						2025.12.1	044						2025.12.1	047						2025.12.1	048						2025.12.1	049						2025.12.1	050						2025.12.1																					
Summa							Techniek College Rotterdam							Zadkine							Technicom							ROC van Twente							OPPStap																											
Piek sessies: 554 Live sessies: 318							Piek sessies: 272 Live sessies: 124							Piek sessies: 607 Live sessies: 336							Piek sessies: 28 Live sessies: 7							Piek sessies: 4 Live sessies: 1							Piek sessies: 128 Live sessies: 1																											
Duur: 0.155s Load average: 0.9							Duur: 0.169s Load average: 0.3							Duur: 0.148s Load average: 0.8							Duur: 0.072s Load average: 0.2							Duur: 0.003s Load average: 0.1							Duur: 0.021s Load average: 0.1																											
Jobs Running: 0 Jobs Queued: 0							Jobs Running: 0 Jobs Queued: 0							Jobs Running: 1 Jobs Queued: 0							Jobs Running: 0 Jobs Queued: 0							Jobs Running: 0 Jobs Queued: 0							Jobs Running: 0 Jobs Queued: 0																											
Geen jobs							Geen jobs							⚠ Workflowautomaat 53%							Geen jobs							Geen jobs							Geen jobs																											
051						2025.12.1																																																								
SG Bonaire																																																														
Piek sessies: 1 Live sessies: 0																																																														
Duur: 0.003s Load average: 0.1																																																														
Jobs Running: 0 Jobs Queued: 0																																																														
Geen jobs																																																														

The "137" Murders



MARTIJN DASHORST

Author Wicket in Action

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 martijn.dashorst@topicus.nl
 martijndashorst.com

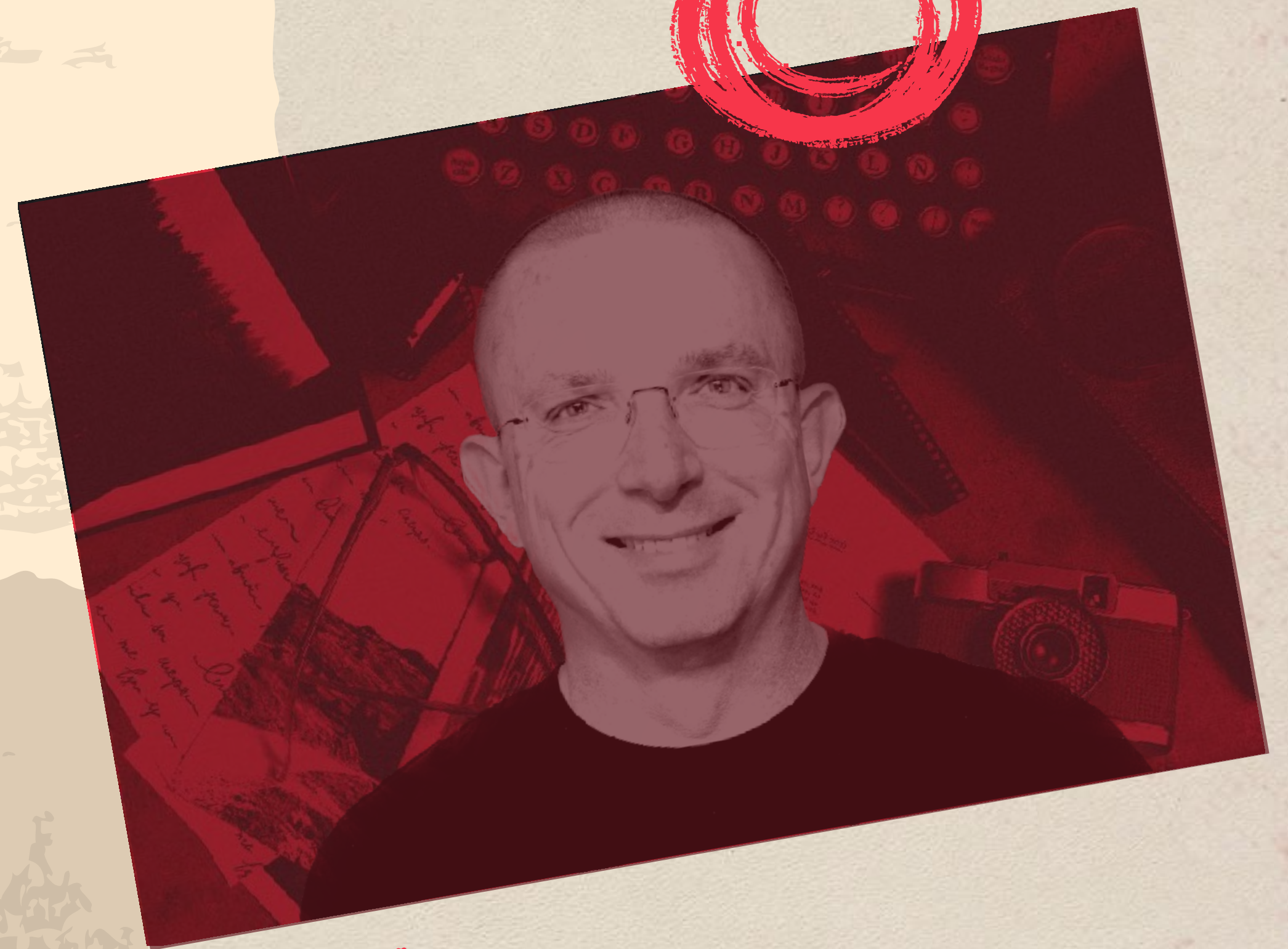


TABLE OF CONTENTS

source?

COLLECTING EVIDENCE

Scavenging the crime scene
for scraps of evidence

CREATING A KILLER PROFILE

To catch a killer you need
to know the killer

ANATOMY OF THE VICTIMS

Why the JVM is a perfect
victim

SURVIVING THE 137-KILLER

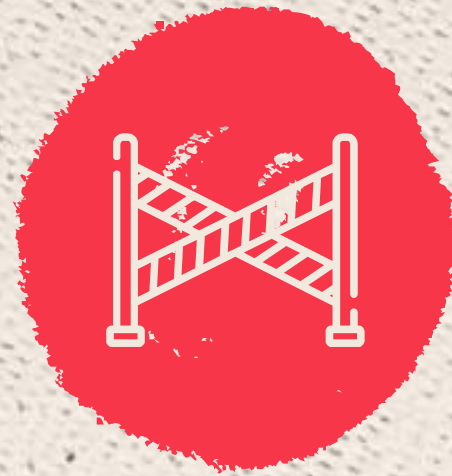
About living a long
container life



COLLECTING THE EVIDENCE

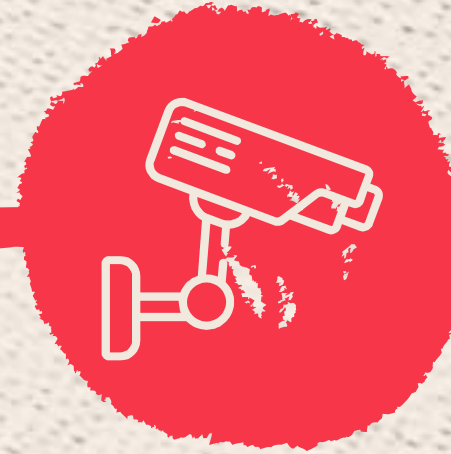
Container Service Investigation

when was?



STEP 1

Check the
crime scene
and
witnesses



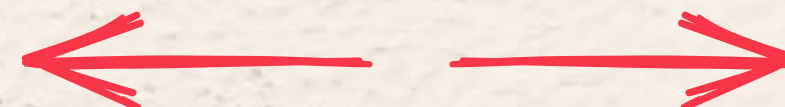
STEP 2

Surveillance
by stake-out
& reviewing
recordings



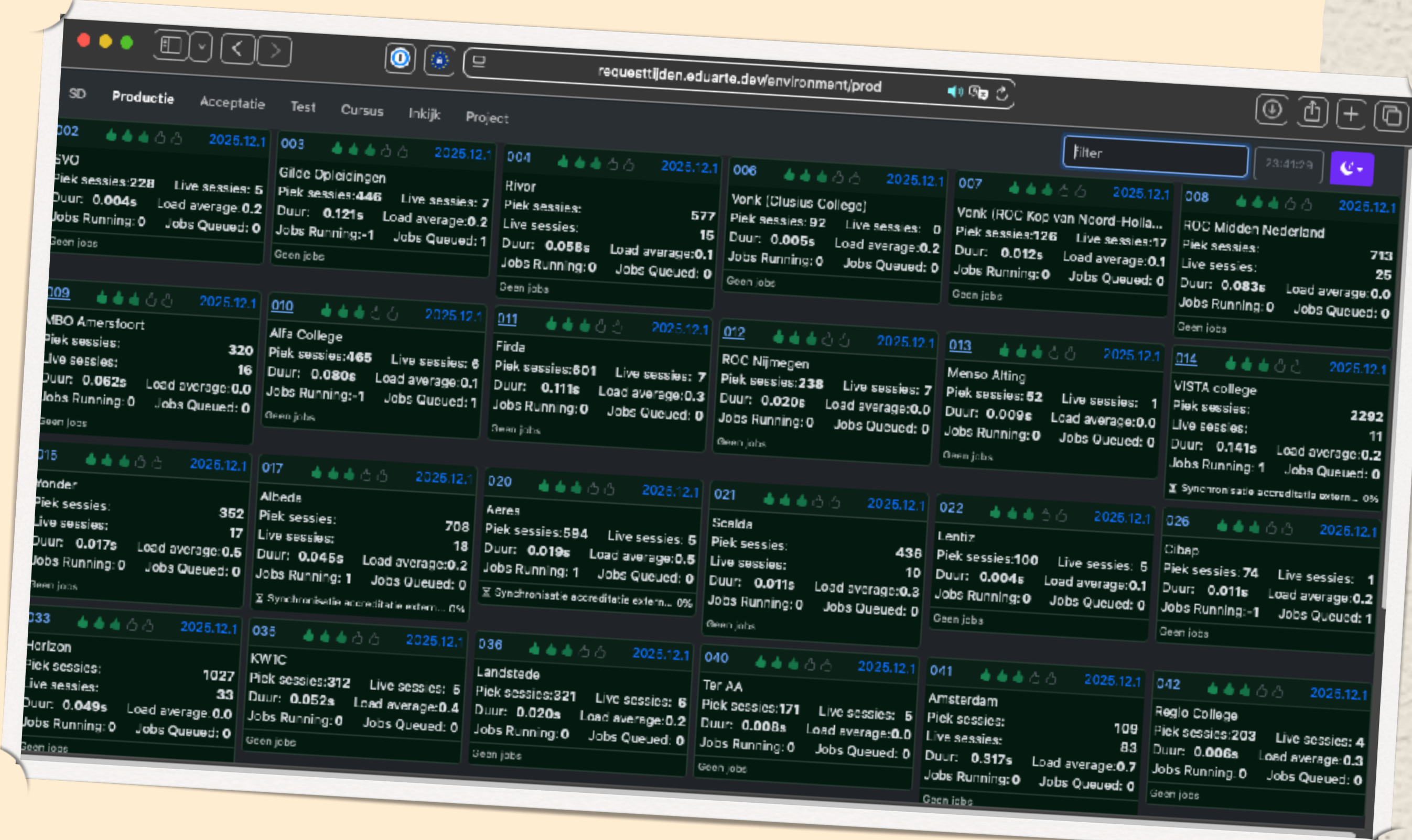
STEP 3

Catalog and
analyse the
gathered
evidence



1. WITNESS REPORT – BRAD S.

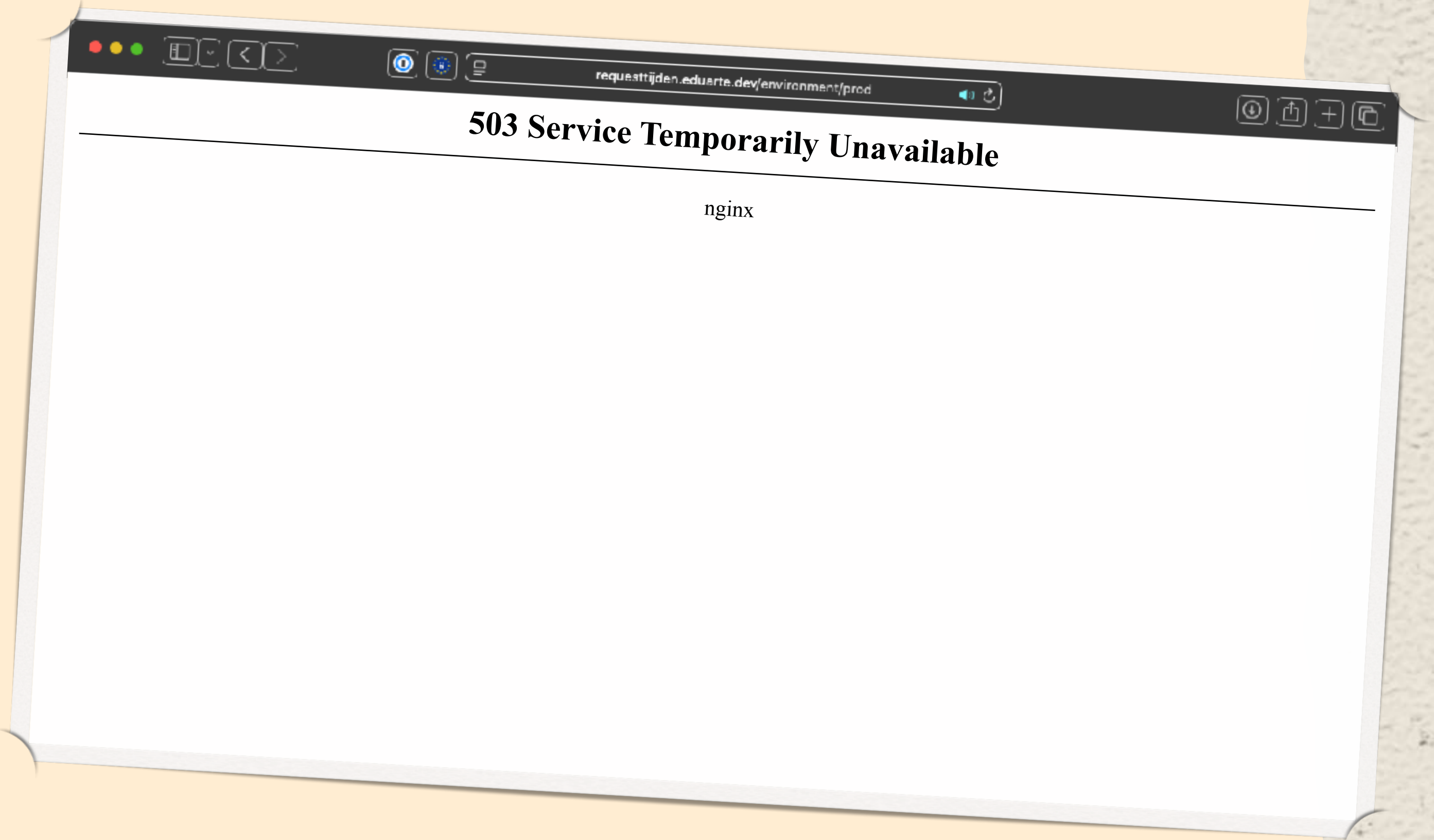
- Brad S. was looking at the requesttijden



BRAD S.

1. WITNESS REPORT – BRAD S.

- Brad S. was looking at the requesttijden
- Gets "503 Service Temporarily Unavailable"



BRAD SMITH

1. WITNESS REPORT – BRAD S.

- Brad S. was looking at the requesttijden
- Gets "503 Service Temporarily Unavailable"
- Application "stopped" working



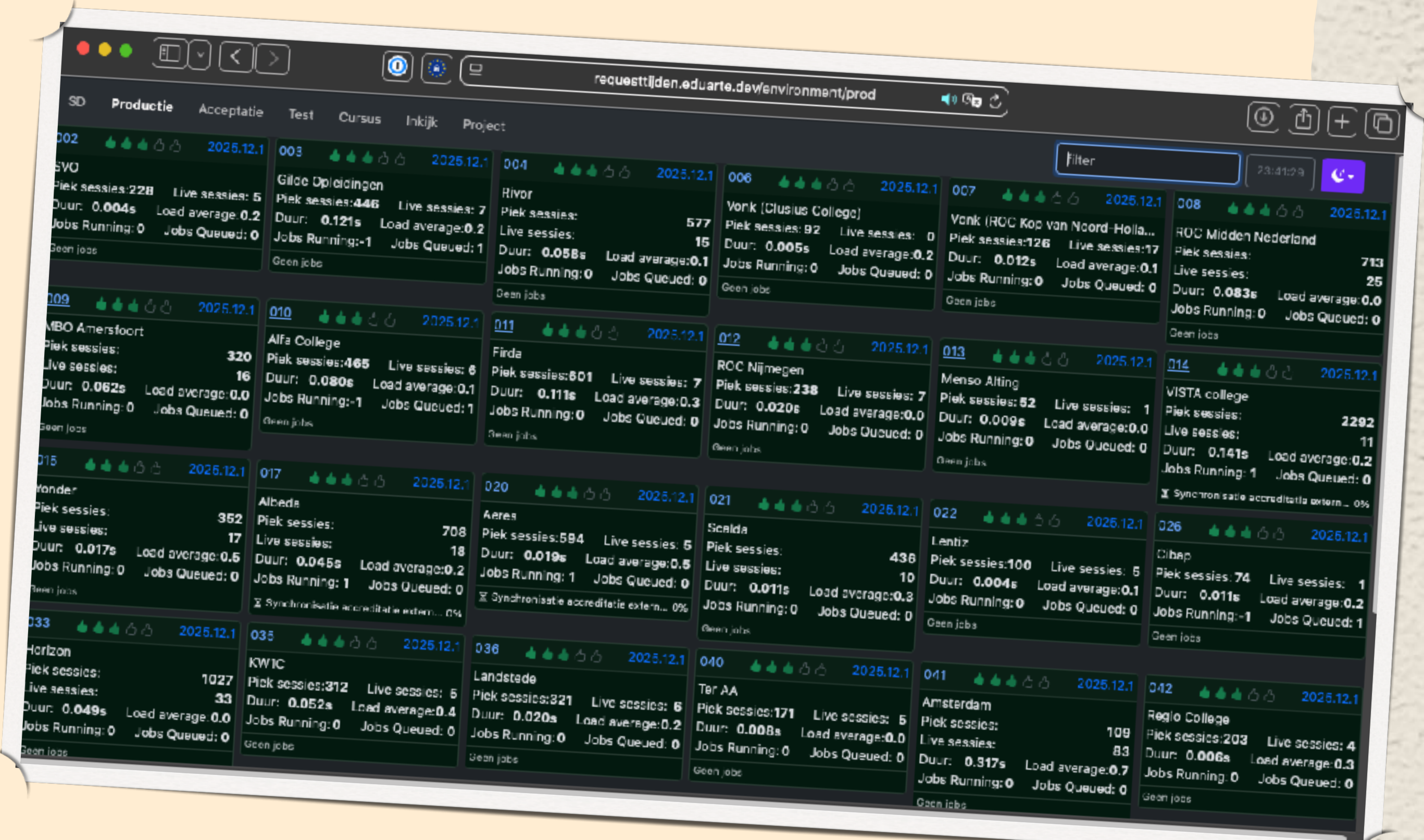
BRAD SMITH

1. WITNESS REPORT – BRAD S.

- Brad S. was looking at the requesttijden
- Gets "503 Service Temporarily Unavailable"
- Application "stopped" working
- Manually refreshing the page makes it work again



BRAD SMITH

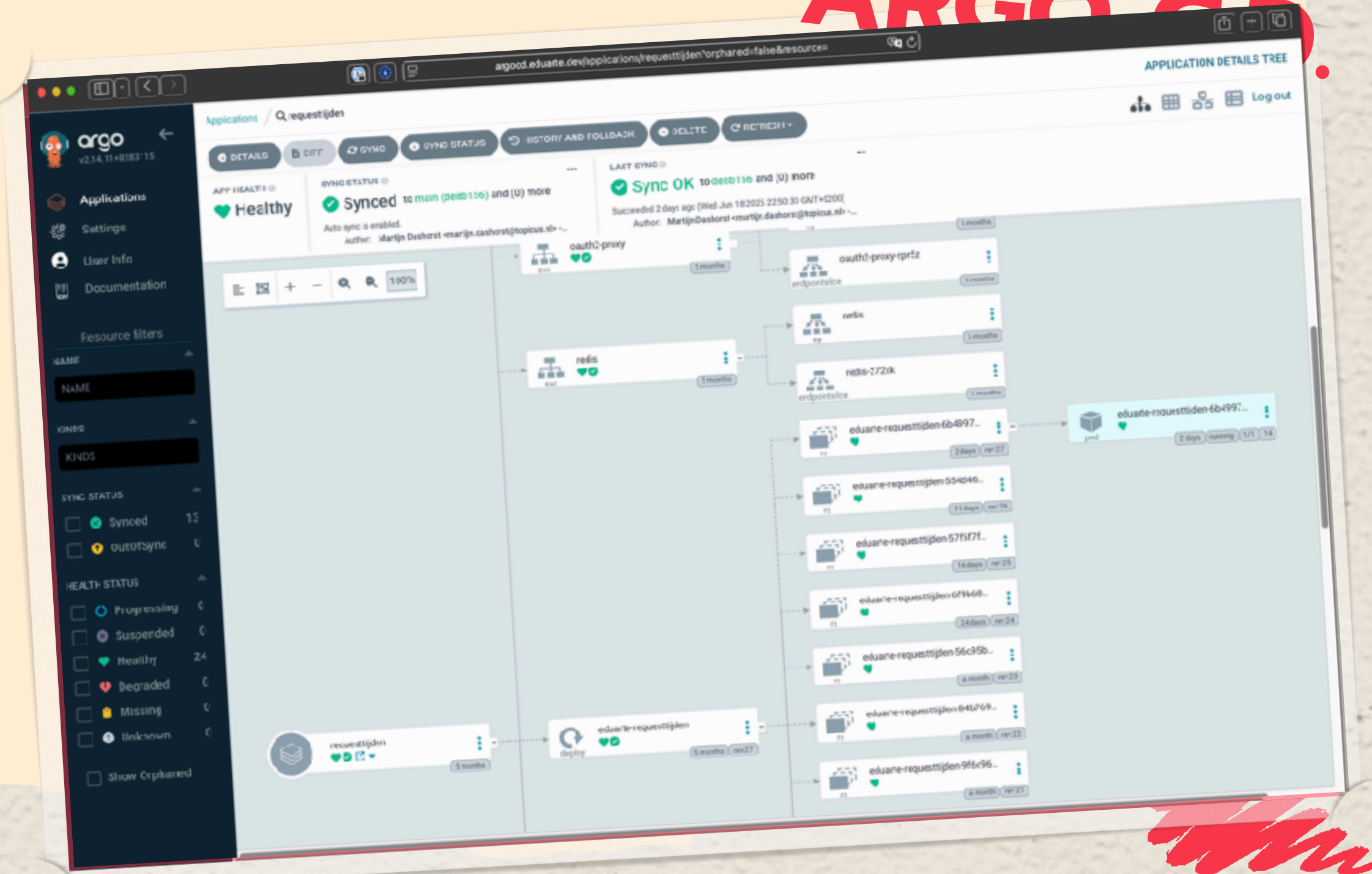


1. WITNESS REPORT – ARGO C.D.

- Argo C.D. works for the temp agency deploying temps to customers in the grander Kubernetes area



ARGO C.D.

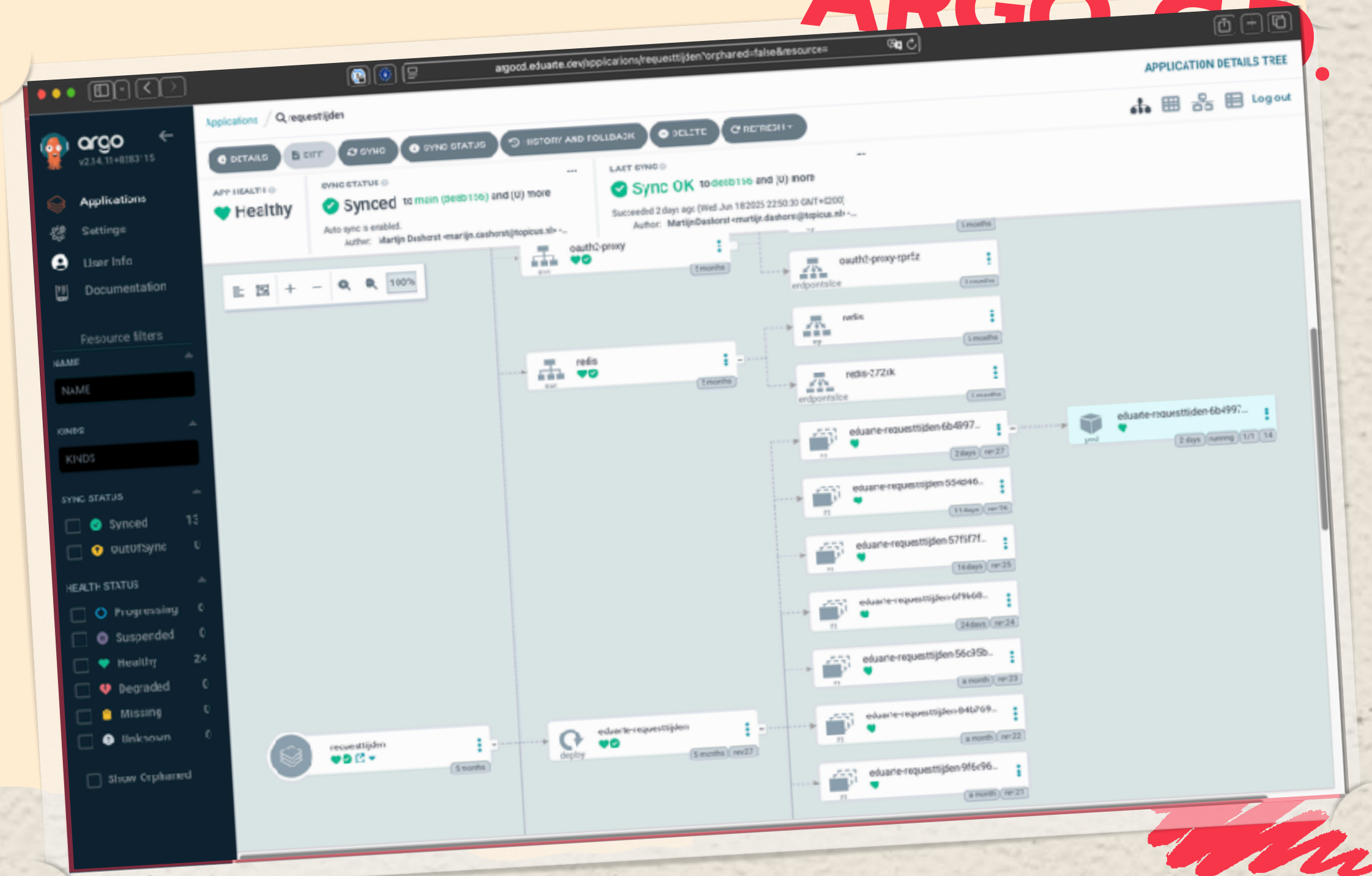


1. WITNESS REPORT – ARGO C.D.

- Argo C.D. works for the temp agency deploying temps to customers in the grander Kubernetes area
- Sent Requesttijden to Kubernetes in Azure



ARGO CD

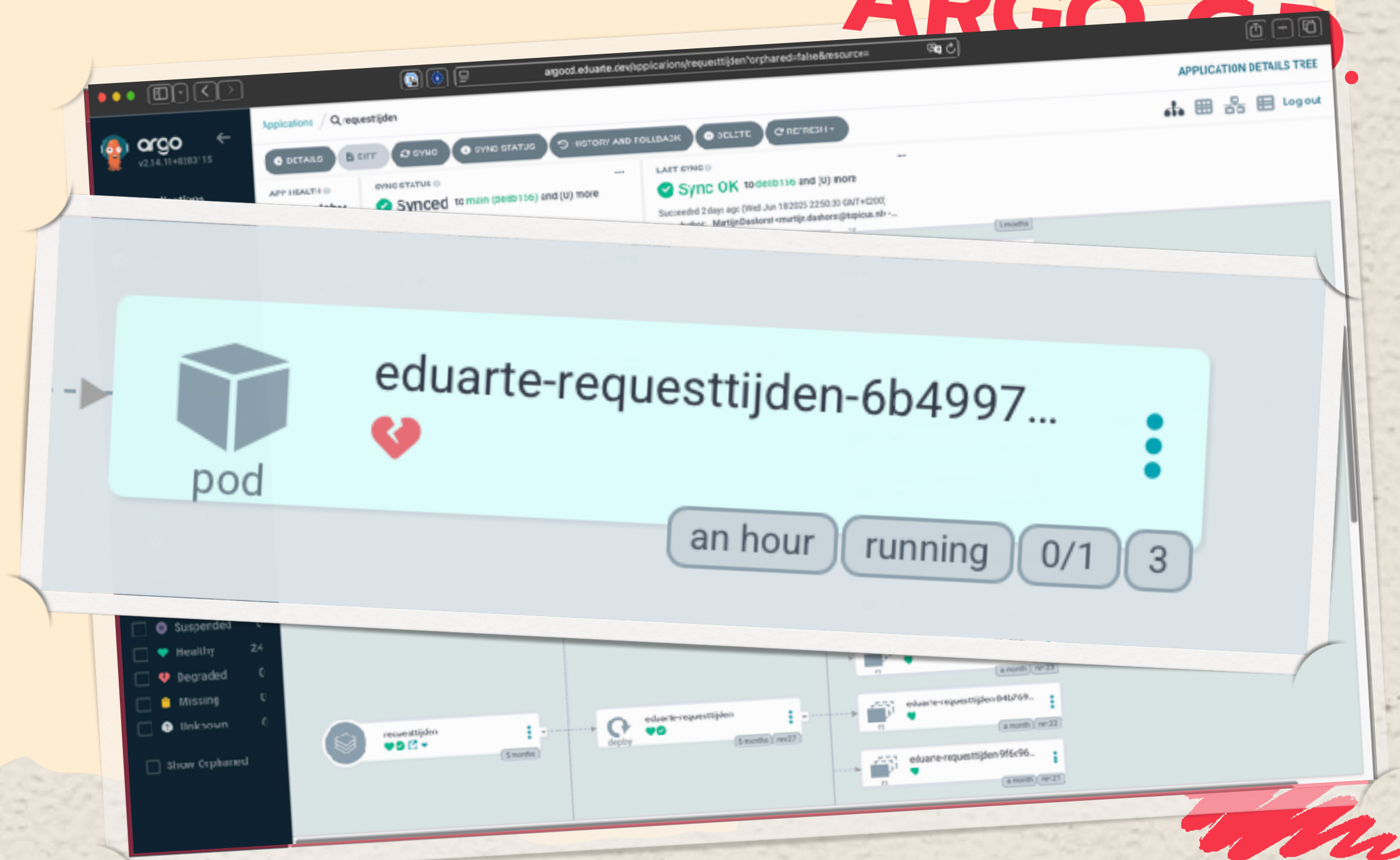


1. WITNESS REPORT – ARGO C.D.

- Argo C.D. works for the temp agency deploying temps to customers in the grander Kubernetes area
- Sent Requesttijden to Kubernetes in Azure
- Noticed the absence of Requesttijden at 11:47am



ARGO C.D.

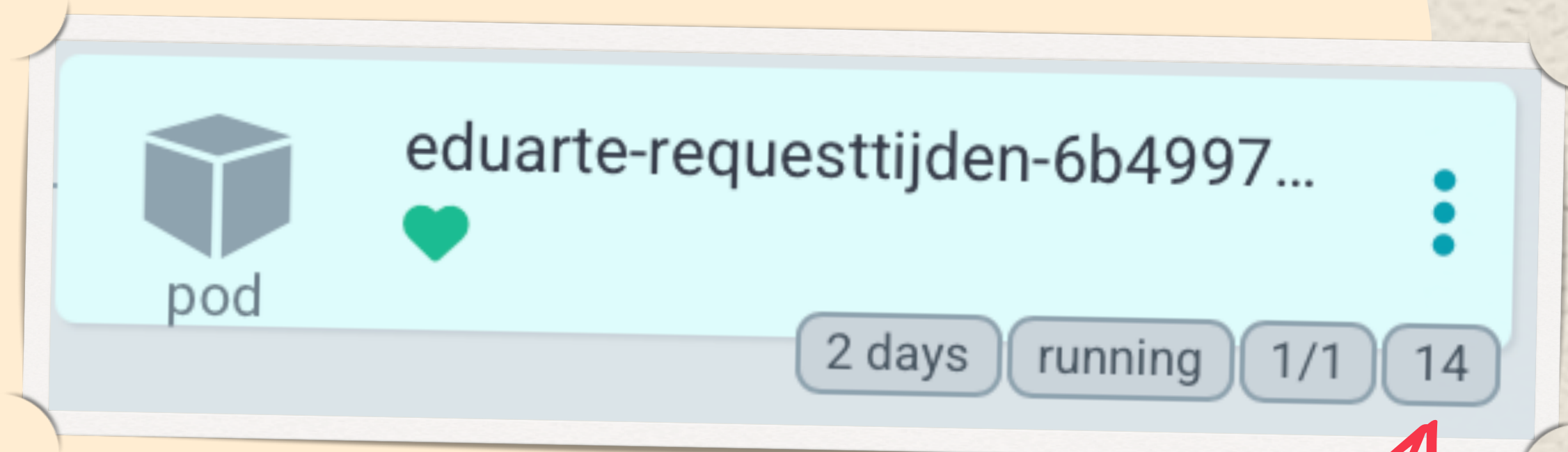


1. WITNESS REPORT – ARGO C.D.

- Argo C.D. works for the temp agency deploying temps to customers in the grander Kubernetes area
- Sent Requesttijden to Kubernetes in Azure
- Noticed the absence of Requesttijden at 11:47am
- Sent a new Requesttijden temp immediately to replace the missing victim



ARGO C.D.



*14 temps in
2 days*



1. WITNESS REPORT – ARGO C.D.

- Argo C.D. works for the temp agency deploying temps to customers in the grander Kubernetes area
- Sent Requesttijden to Kubernetes in Azure
- Noticed the absence of Requesttijden at 11:47am
- Sent a new Requesttijden temp immediately to replace the missing victim
- Noticed a note with 137 on it and the term "OOMKilled"



ARGO C.D.

*The container last terminated
12 hours ago with exit code
137 because of OOMKilled.*



1. THE CRIME SCENE

- The console output just stopped



1. THE CRIME SCENE

The console output just stopped



1. THE CRIME SCENE

- The console output just stopped
- No indication of stopping

2025-06-18 21:20:37,831 INFO [req.cra.Crawler]
(executor-thread-39) Completed crawling 144 status
pages (failed: 3) #61 in 4.83s



1. THE CRIME SCENE

- The console output just stopped
- No indication of stopping
- Liveness probe failed with 503

Liveness probe failed: HTTP probe failed with statuscode: 503



Unhealthy

Liveness probe failed: HTTP probe failed with
statuscode: 503

1

2m ago
Today at 11:21 PM

2m ago
Today at 11:21 PM

1. THE CRIME SCENE

- The console output just stopped
- No indication of stopping
- Liveness probe failed with 503
- Kubernetes shows event "Error (OOMKilled)" and "Exit Code 137"

requesttijden	
Status	Running
Started	18/06/2025, 23:21:08 CEST
Restart Count	1
Last State	Error (OOMKilled) Exit Code 137 Started 18/06/2025, 22:50:31 CEST Finished 18/06/2025, 23:21:07 CEST
Container ID	containerd://c369d16b8b75e65f04cfbe4f8ae83eef62bb400b49228908e622079397c15f33
Image Pull Policy	Always
Image	eduardedev.azurecr.io/eduarde-requesttijden2:1.0.10

Last State

Error (OOMKilled)

Exit Code 137

1. THE CRIME SCENE

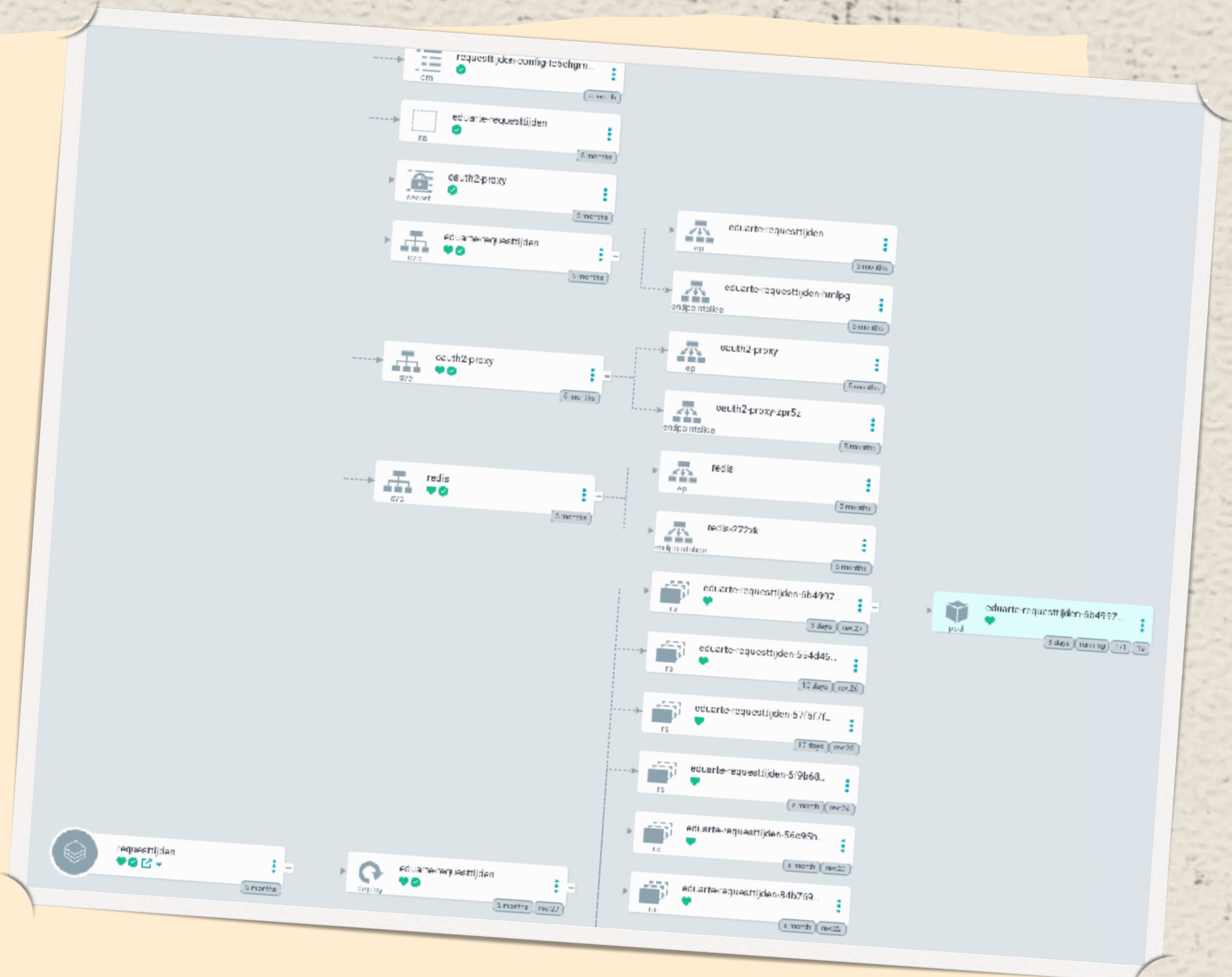
- The console output just stopped
- No indication of stopping
- Readiness probe failed with 503
- Kubernetes shows event "137 OOMKilled"
- Java probes can't find JVM process

```
> kubectl exec -it eduarte-requesttijden-6b4997c7df-v5gvn -- jcmd 1 VM.uptime
1:
java.io.IOException: No such process
    at jdk.attach/sun.tools.attach.VirtualMachineImpl.sendQuitTo(Native
command terminated with exit code 1
```

```
🍏 > 📁 ~/Workspaces/Eduarte.JEE8/requesttijden > on 🐱 🧑 main *2 ?2 .....
```


2. SURVEILLANCE – STAKE OUT

- Watching console output
- Watching Argo C.D.'s layout



2. SURVEILLANCE – STAKE OUT

- Watching console output
- Watching Argo C.D.'s layout
- Watching Kubernetes Console (lens, headlamp)



2. SURVEILLANCE – STAKE OUT

- Watching console output
- Watching Argo C.D.'s layout
- Watching Kubernetes Console (lens, headlamp)

No results



2. SURVEILLANCE – GRAFANA

CPU USAGE

Node, container,
JVM, etc

STATUSESSES

restarts, exit
code, etc.

MEM USAGE

RSS, container,
JVM, (non-)heap

#THREADS

Container, JVM

TIME TRAVEL

Go back to restarts



SNIPPED DUE TO FILE SIZE CONSTRAINTS

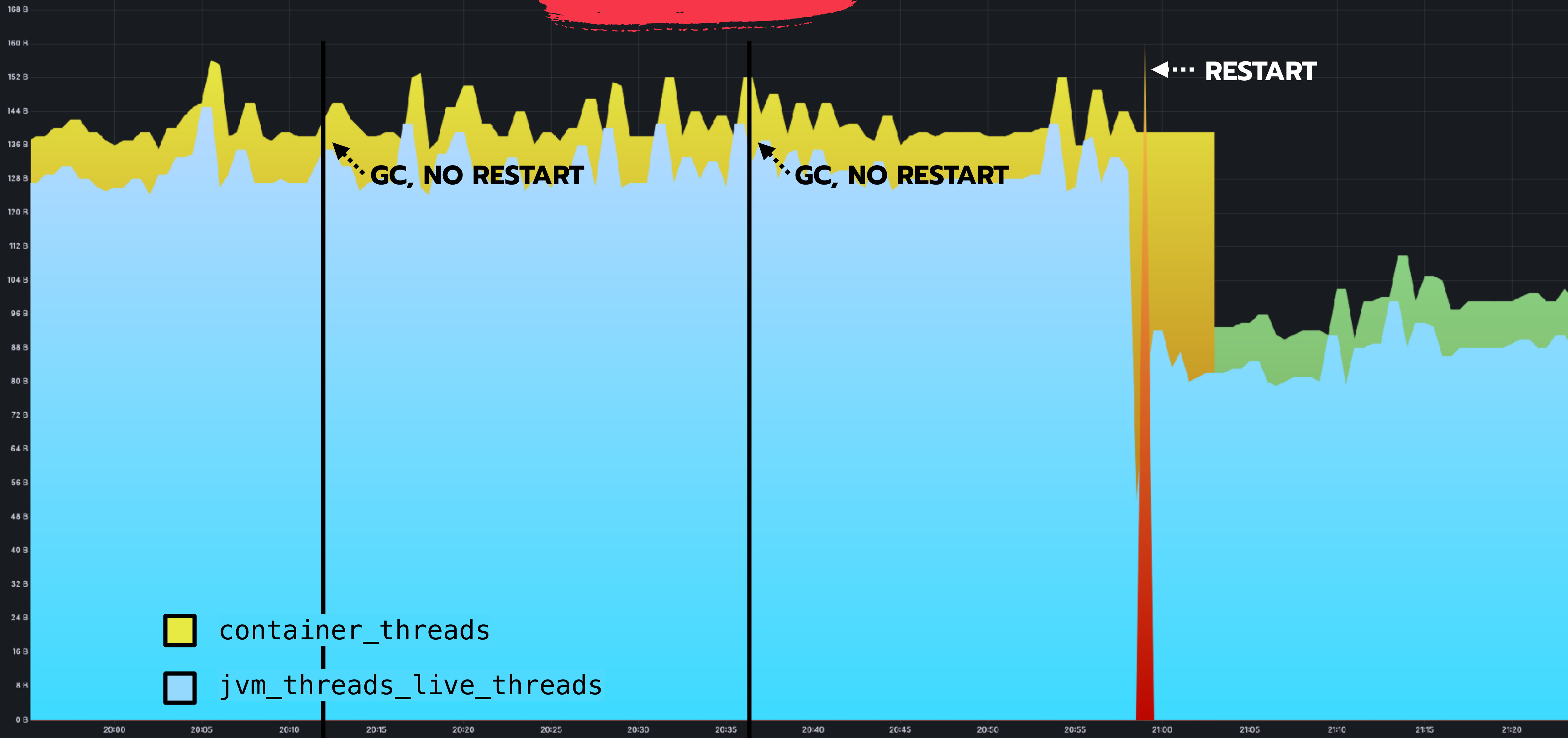
XXXX

~~~~~



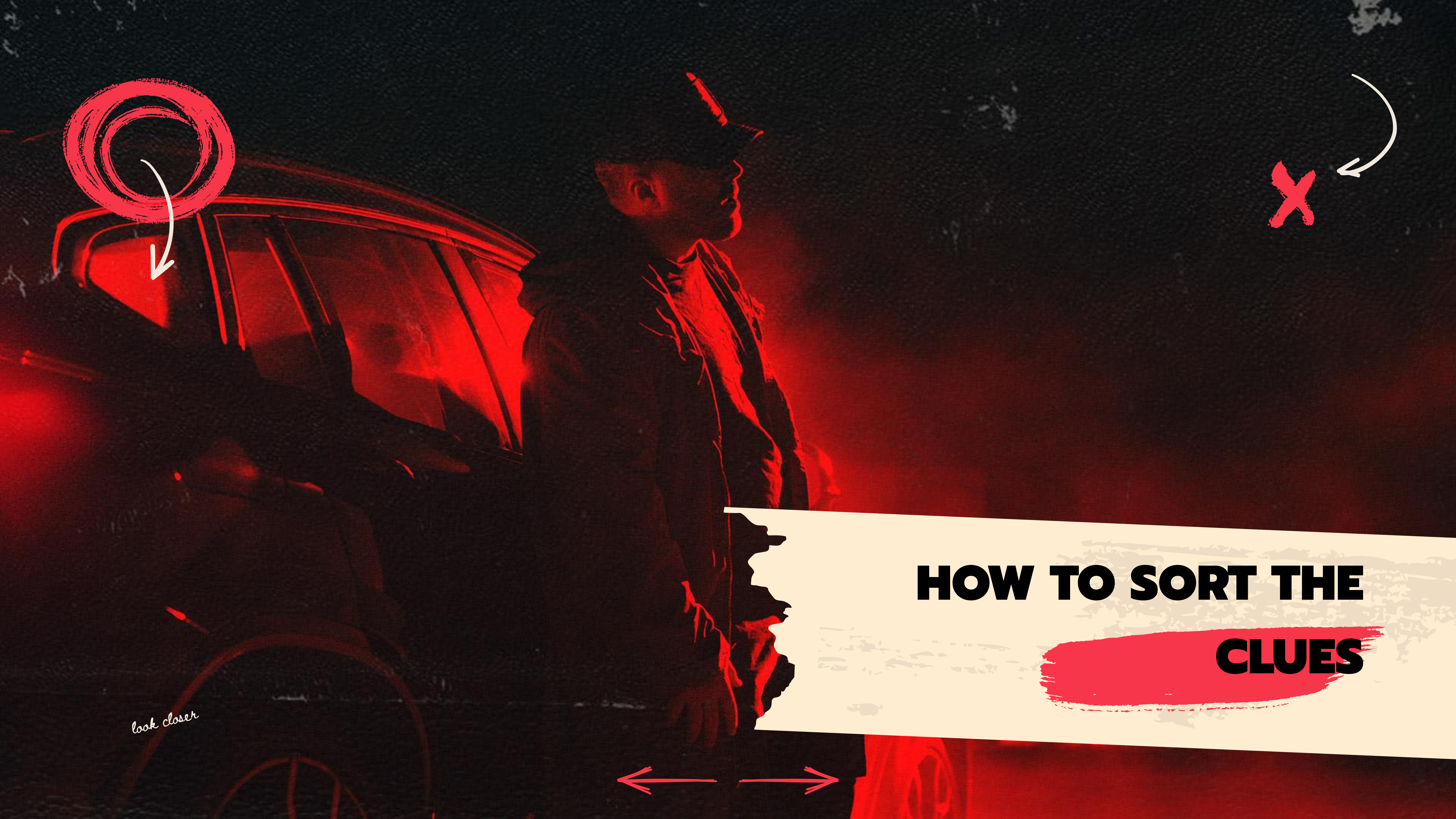
All threads

# CONTAINER & JVM THREAD USAGE



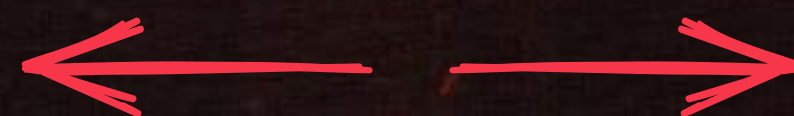
- container\_threads
- jvm\_threads\_live\_threads





look closer

# HOW TO SORT THE CLUES





### 3. THE EVIDENCE

**WHERE?**

In kubernetes

**WHEN?**

High container  
memory use,  
random intervals

**WHAT?**

Java app

**HOW?**

Code **137**  
OOMKilled

**WHO?**

"The 137 Killer"  
aka OOMKiller



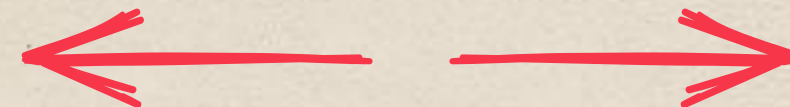


# THE SIGNIFICANCE OF 137 & OOMKILLED

*Signature?*

Container signals are codes the system uses to control processes, often shown as an exit code.

- 0 := SUCCESS | NORMAL EXIT
- 128 := externally triggered
- 137 = 128 + 9
- 9 := SIG\_KILL





# 137 = Out Of Memory?

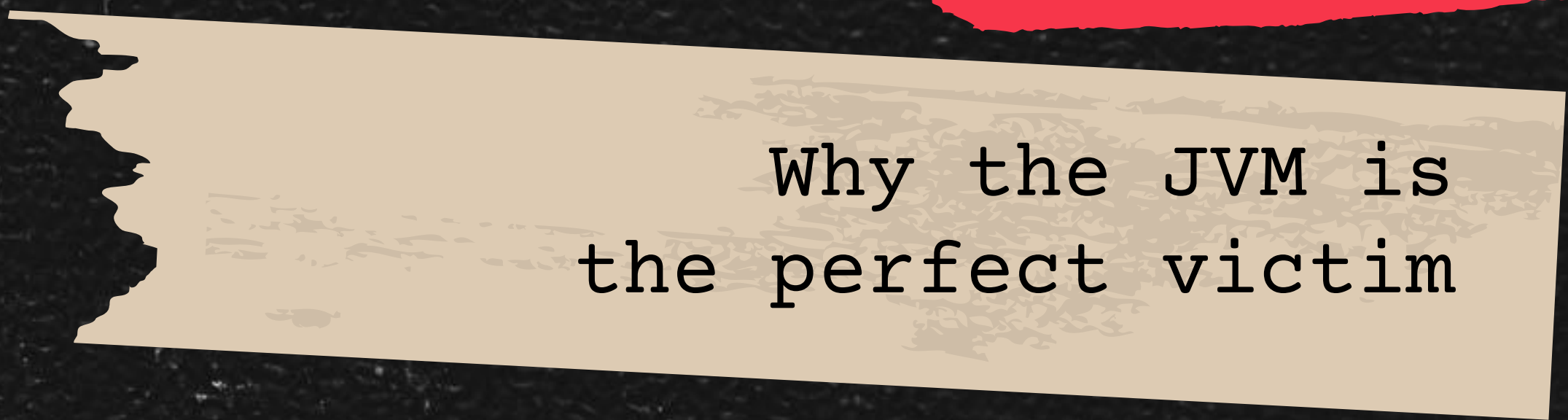
The most significant clue  
for our investigation!







# ANATOMY OF THE VICTIMS



Why the JVM is  
the perfect victim







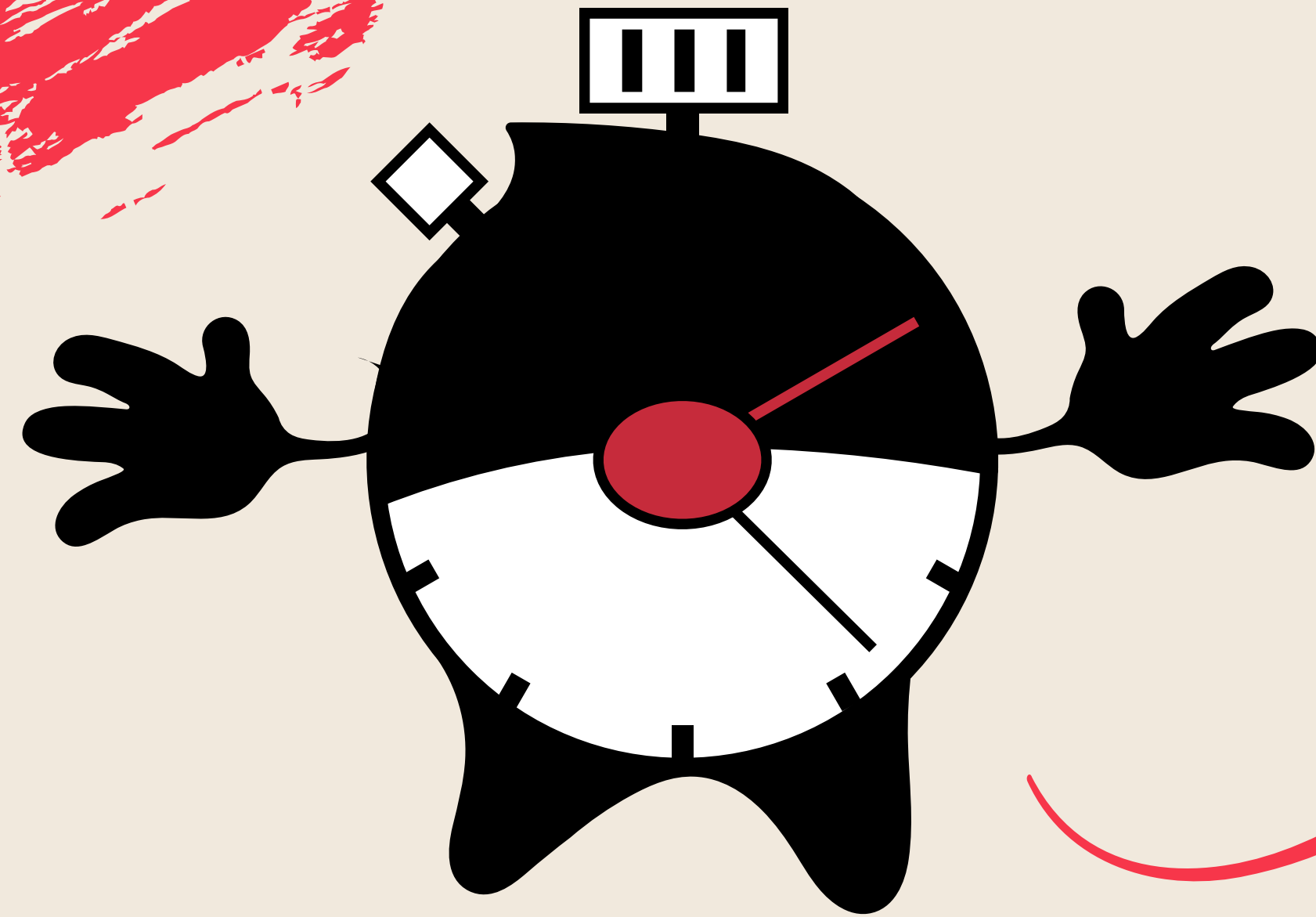
XXXXX

# REQUESTTIJDEN

- RETRIEVES STATUS  
PAGES EVERY 30 SECS  
FOR ~150 SERVERS
- ~10 CONCURRENT USERS
- RUNS IN KUBERNETES







edo.educus.nl/app/status

Eduarte statuspagina - Algemeen

algemeenjvmservletscheduler

Systeem

Starttijd

CPU cores

Load average

Java VM

Java vendor

Hostname

06-06-2025 10:47

16

0,01

OpenJDK 64-Bit Server VM

Azul Systems, Inc.

d-wfy092a.eduarteaad.nl

Geheugen

Maximum geheugen

Gebruikt geheugen

Vrij geheugen

8GB

3,1GB

4,9GB

Applicatie

Versie

Registered pages

Build

Status

2024.9.2, sha1=25af40d7828

0

onbekend

OK

Nr Errors

Pagina requests

Achtergrondtaken

0 (1)

0

Scheduler

Jobs Completed (total)

Jobs Running (total)

Jobs Queued (total)

5767

0

0

Wicket Sessies/Requests

Piek sessies

Live sessies

Actieve requests

Gem. request duur

Requests per minuut

2

2

1

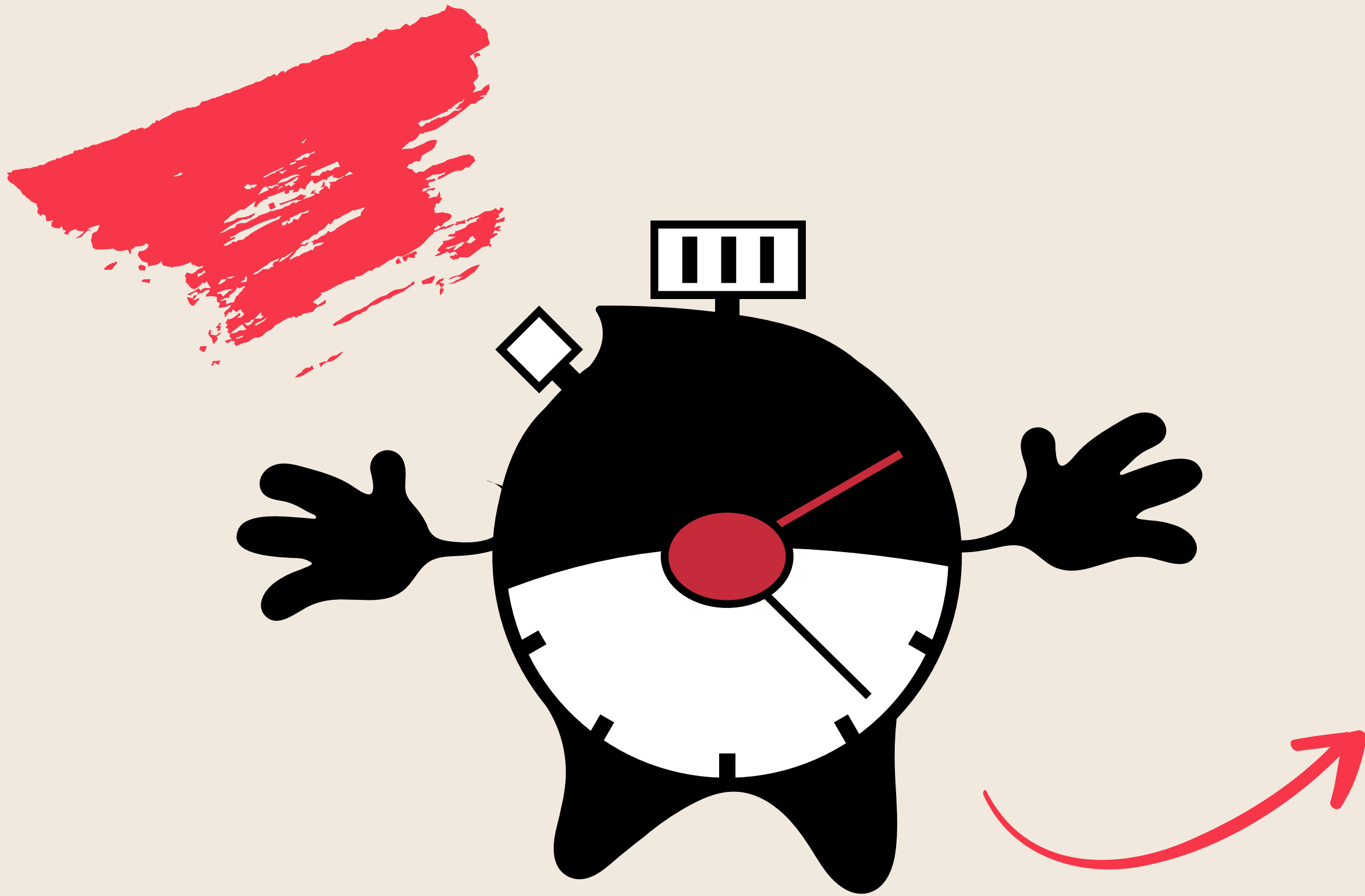
4 ms.

8

# REQUESTTIJDEN

EVERY 30 SECONDS  
RETRIEVES STATUS PAGES  
FOR ~150 SERVERS





Eduarte statuspagina - Algemeen

[algemeen](#) [jvm](#) [servlet](#) [scheduler](#)

Systeem

Starttijd06-06-2025 10:47

CPU cores16

Load average0,01

Java VMOpenJDK 64-Bit Server VM

Java vendorAzul Systems, Inc.

Hostnamed-wfy092a.eduarteaad.nl

Geheugen

Maximum geheugen8GB

Gebruikt geheugen3,1GB

Vrij geheugen4,9GB

Applicatie

Versie2024.9.2, sha1=25af40d7828

Registered pages0

Buildonbekend

StatusOK

Nr Errors

Pagina requests0 (1)

Achtergrondtaken0

Scheduler

Jobs Completed (total)5767

Jobs Running (total)0

Jobs Queued (total)0

Wicket Sessies/Requests

Piek sessies2

Live sessies2

Actieve requests1

Gem. request duur4 ms.

Requests per minuut8

SDO

2024.9.2

SDO

Piek sessies:

3

Live sessies:

0

Duur:

0.004s

Load average:

0.3

Jobs Running:

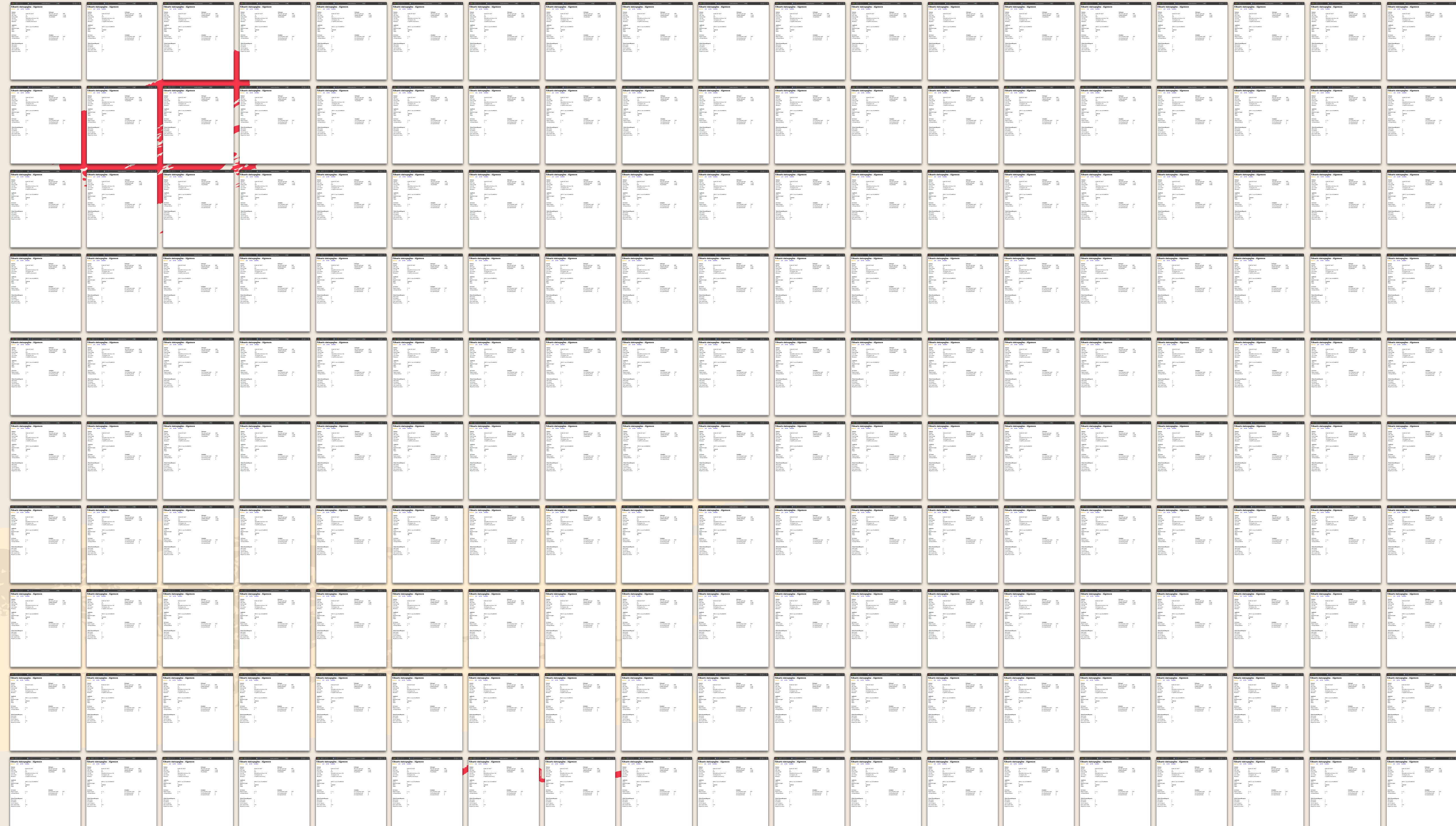
0

Jobs Queued:

0

Geen jobs







SD

Productie

Acceptatie

Test

Cursus

Inkijk

Project

filter

18:08:58

|                     |                   |             |             |             |             |           |                     |                   |             |             |             |             |           |                     |                   |             |             |             |             |           |                        |                   |             |             |             |             |                                  |                     |             |             |             |             |             |                      |                     |             |             |             |             |             |           |
|---------------------|-------------------|-------------|-------------|-------------|-------------|-----------|---------------------|-------------------|-------------|-------------|-------------|-------------|-----------|---------------------|-------------------|-------------|-------------|-------------|-------------|-----------|------------------------|-------------------|-------------|-------------|-------------|-------------|----------------------------------|---------------------|-------------|-------------|-------------|-------------|-------------|----------------------|---------------------|-------------|-------------|-------------|-------------|-------------|-----------|
| <a href="#">002</a> | <div></div>       | <div></div> | <div></div> | <div></div> | <div></div> | 2025.12.1 | <a href="#">003</a> | <div></div>       | <div></div> | <div></div> | <div></div> | <div></div> | 2025.12.1 | <a href="#">004</a> | <div></div>       | <div></div> | <div></div> | <div></div> | <div></div> | 2025.12.1 | <a href="#">006</a>    | <div></div>       | <div></div> | <div></div> | <div></div> | <div></div> | 2025.12.1                        | <a href="#">007</a> | <div></div> | <div></div> | <div></div> | <div></div> | <div></div> | 2025.12.1            | <a href="#">008</a> | <div></div> | <div></div> | <div></div> | <div></div> | <div></div> | 2025.12.1 |
| SVO                 |                   |             |             |             |             |           | Gilde Opleidingen   |                   |             |             |             |             |           | Rivor               |                   |             |             |             |             |           | Vonk (Clusius College) |                   |             |             |             |             | Vonk (ROC Kop van Noord-Holland) |                     |             |             |             |             |             | ROC Midden Nederland |                     |             |             |             |             |             |           |
| Piek sessies: 228   | Live sessies: 0   |             |             |             |             |           | Piek sessies: 431   | Live sessies: 6   |             |             |             |             |           | Piek sessies: 577   | Live sessies: 16  |             |             |             |             |           | Piek sessies: 92       | Live sessies: 0   |             |             |             |             | Piek sessies: 126                | Live sessies: 16    |             |             |             |             |             | Piek sessies: 533    | Live sessies: 43    |             |             |             |             |             |           |
| Duur: 0.003s        | Load average: 0.1 |             |             |             |             |           | Duur: 0.099s        | Load average: 0.0 |             |             |             |             |           | Duur: 0.015s        | Load average: 0.0 |             |             |             |             |           | Duur: 0.004s           | Load average: 0.0 |             |             |             |             | Duur: 0.013s                     | Load average: 0.1   |             |             |             |             |             | Duur: 0.167s         | Load average: 0.3   |             |             |             |             |             |           |
| Jobs Running: 0     | Jobs Queued: 0    |             |             |             |             |           | Jobs Running: -1    | Jobs Queued: 1    |             |             |             |             |           | Jobs Running: 0     | Jobs Queued: 0    |             |             |             |             |           | Jobs Running: 0        | Jobs Queued: 0    |             |             |             |             | Jobs Running: 0                  | Jobs Queued: 0      |             |             |             |             |             | Jobs Running: 0      | Jobs Queued: 0      |             |             |             |             |             |           |
| Geen jobs           |                   |             |             |             |             |           | Geen jobs           |                   |             |             |             |             |           | Geen jobs           |                   |             |             |             |             |           | Geen jobs              |                   |             |             |             |             | Geen jobs                        |                     |             |             |             |             |             | Geen jobs            |                     |             |             |             |             |             |           |

|                     |                   |             |             |             |             |           |                     |                   |             |             |             |             |           |                     |                   |             |             |             |             |           |                     |                   |             |             |             |             |                  |                     |             |             |             |             |             |                   |                     |             |             |             |             |             |           |
|---------------------|-------------------|-------------|-------------|-------------|-------------|-----------|---------------------|-------------------|-------------|-------------|-------------|-------------|-----------|---------------------|-------------------|-------------|-------------|-------------|-------------|-----------|---------------------|-------------------|-------------|-------------|-------------|-------------|------------------|---------------------|-------------|-------------|-------------|-------------|-------------|-------------------|---------------------|-------------|-------------|-------------|-------------|-------------|-----------|
| <a href="#">009</a> | <div></div>       | <div></div> | <div></div> | <div></div> | <div></div> | 2025.12.1 | <a href="#">010</a> | <div></div>       | <div></div> | <div></div> | <div></div> | <div></div> | 2025.12.1 | <a href="#">011</a> | <div></div>       | <div></div> | <div></div> | <div></div> | <div></div> | 2025.12.1 | <a href="#">012</a> | <div></div>       | <div></div> | <div></div> | <div></div> | <div></div> | 2025.12.1        | <a href="#">013</a> | <div></div> | <div></div> | <div></div> | <div></div> | <div></div> | 2025.12.1         | <a href="#">014</a> | <div></div> | <div></div> | <div></div> | <div></div> | <div></div> | 2025.12.1 |
| MBO Amersfoort      |                   |             |             |             |             |           | Alfa College        |                   |             |             |             |             |           | Firda               |                   |             |             |             |             |           | ROC Nijmegen        |                   |             |             |             |             | Menso Alting     |                     |             |             |             |             |             | VISTA college     |                     |             |             |             |             |             |           |
| Piek sessies: 262   | Live sessies: 18  |             |             |             |             |           | Piek sessies: 465   | Live sessies: 16  |             |             |             |             |           | Piek sessies: 601   | Live sessies: 17  |             |             |             |             |           | Piek sessies: 229   | Live sessies: 3   |             |             |             |             | Piek sessies: 52 | Live sessies: 1     |             |             |             |             |             | Piek sessies: 381 | Live sessies: 9     |             |             |             |             |             |           |
| Duur: 0.016s        | Load average: 0.0 |             |             |             |             |           | Duur: 0.024s        | Load average: 0.2 |             |             |             |             |           | Duur: 0.143s        | Load average: 0.0 |             |             |             |             |           | Duur: 0.032s        | Load average: 0.9 |             |             |             |             | Duur: 0.004s     | Load average: 0.1   |             |             |             |             |             | Duur: 0.179s      | Load average: 0.1   |             |             |             |             |             |           |
| Jobs Running: 0     | Jobs Queued: 0    |             |             |             |             |           | Jobs Running: 0     | Jobs Queued: 0    |             |             |             |             |           | Jobs Running: 0     | Jobs Queued: 0    |             |             |             |             |           | Jobs Running: 1     | Jobs Queued: 0    |             |             |             |             | Jobs Running: 0  | Jobs Queued: 0      |             |             |             |             |             | Jobs Running: 0   | Jobs Queued: 0      |             |             |             |             |             |           |
| Geen jobs           |                   |             |             |             |             |           | Geen jobs           |                   |             |             |             |             |           | Geen jobs           |                   |             |             |             |             |           | 🔄 Workflowautomaat  | 24%               |             |             |             |             | Geen jobs        |                     |             |             |             |             |             | Geen jobs         |                     |             |             |             |             |             |           |

|                     |                   |             |             |             |             |           |                     |                   |             |             |             |             |           |                     |                   |             |             |             |             |           |                     |                   |             |             |             |             |                   |                     |             |             |             |             |             |                  |                     |             |             |             |             |             |           |
|---------------------|-------------------|-------------|-------------|-------------|-------------|-----------|---------------------|-------------------|-------------|-------------|-------------|-------------|-----------|---------------------|-------------------|-------------|-------------|-------------|-------------|-----------|---------------------|-------------------|-------------|-------------|-------------|-------------|-------------------|---------------------|-------------|-------------|-------------|-------------|-------------|------------------|---------------------|-------------|-------------|-------------|-------------|-------------|-----------|
| <a href="#">015</a> | <div></div>       | <div></div> | <div></div> | <div></div> | <div></div> | 2025.12.1 | <a href="#">017</a> | <div></div>       | <div></div> | <div></div> | <div></div> | <div></div> | 2025.12.1 | <a href="#">020</a> | <div></div>       | <div></div> | <div></div> | <div></div> | <div></div> | 2025.12.1 | <a href="#">021</a> | <div></div>       | <div></div> | <div></div> | <div></div> | <div></div> | 2025.12.1         | <a href="#">022</a> | <div></div> | <div></div> | <div></div> | <div></div> | <div></div> | 2025.12.1        | <a href="#">026</a> | <div></div> | <div></div> | <div></div> | <div></div> | <div></div> | 2025.12.1 |
| Yonder              |                   |             |             |             |             |           | Albeda              |                   |             |             |             |             |           | Aeres               |                   |             |             |             |             |           | Scalda              |                   |             |             |             |             | Lentiz            |                     |             |             |             |             |             | Cibap            |                     |             |             |             |             |             |           |
| Piek sessies: 352   | Live sessies: 17  |             |             |             |             |           | Piek sessies: 702   | Live sessies: 15  |             |             |             |             |           | Piek sessies: 153   | Live sessies: 4   |             |             |             |             |           | Piek sessies: 438   | Live sessies: 2   |             |             |             |             | Piek sessies: 100 | Live sessies: 3     |             |             |             |             |             | Piek sessies: 65 | Live sessies: 3     |             |             |             |             |             |           |
| Duur: 0.028s        | Load average: 1.1 |             |             |             |             |           | Duur: 0.046s        | Load average: 0.1 |             |             |             |             |           | Duur: 0.024s        | Load average: 0.2 |             |             |             |             |           | Duur: 0.053s        | Load average: 0.1 |             |             |             |             | Duur: 0.027s      | Load average: 0.1   |             |             |             |             |             | Duur: 0.007s     | Load average: 0.0   |             |             |             |             |             |           |
| Jobs Running: 1     | Jobs Queued: 0    |             |             |             |             |           | Jobs Running: 0     | Jobs Queued: 0    |             |             |             |             |           | Jobs Running: 1     | Jobs Queued: 0    |             |             |             |             |           | Jobs Running: 0     | Jobs Queued: 0    |             |             |             |             | Jobs Running: 0   | Jobs Queued: 0      |             |             |             |             |             | Jobs Running: -1 | Jobs Queued: 1      |             |             |             |             |             |           |
| Geen jobs           |                   |             |             |             |             |           | Geen jobs           |                   |             |             |             |             |           | 🔄 Workflowautomaat  | 8%                |             |             |             |             |           | Geen jobs           |                   |             |             |             |             | Geen jobs         |                     |             |             |             |             |             | Geen jobs        |                     |             |             |             |             |             |           |

|                       |                   |             |             |             |             |           |                     |                   |             |             |             |             |           |                     |                   |             |             |             |             |           |                     |                   |             |             |             |             |                    |                     |             |             |             |             |             |                   |                     |             |             |             |             |             |           |
|-----------------------|-------------------|-------------|-------------|-------------|-------------|-----------|---------------------|-------------------|-------------|-------------|-------------|-------------|-----------|---------------------|-------------------|-------------|-------------|-------------|-------------|-----------|---------------------|-------------------|-------------|-------------|-------------|-------------|--------------------|---------------------|-------------|-------------|-------------|-------------|-------------|-------------------|---------------------|-------------|-------------|-------------|-------------|-------------|-----------|
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| Horizon               |                   |             |             |             |             |           | KW1C                |                   |             |             |             |             |           | Landstede           |                   |             |             |             |             |           | Ter AA              |                   |             |             |             |             | Amsterdam          |                     |             |             |             |             |             | Regio College     |                     |             |             |             |             |             |           |
| Piek sessies: 482     | Live sessies: 8   |             |             |             |             |           | Piek sessies: 298   | Live sessies: 6   |             |             |             |             |           | Piek sessies: 254   | Live sessies: 7   |             |             |             |             |           | Piek sessies: 149   | Live sessies: 3   |             |             |             |             | Piek sessies: 2093 | Live sessies: 64    |             |             |             |             |             | Piek sessies: 203 | Live sessies: 5     |             |             |             |             |             |           |
| Duur: 0.007s          | Load average: 0.4 |             |             |             |             |           | Duur: 0.228s        | Load average: 0.1 |             |             |             |             |           | Duur: 0.034s        | Load average: 0.0 |             |             |             |             |           | Duur: 0.030s        | Load average: 0.0 |             |             |             |             | Duur: 0.248s       | Load average: 0.6   |             |             |             |             |             | Duur: 0.004s      | Load average: 0.1   |             |             |             |             |             |           |
| Jobs Running: 1       | Jobs Queued: 0    |             |             |             |             |           | Jobs Running: 0     | Jobs Queued: 0    |             |             |             |             |           | Jobs Running: 0     | Jobs Queued: 0    |             |             |             |             |           | Jobs Running: 0     | Jobs Queued: 0    |             |             |             |             | Jobs Running: -1   | Jobs Queued: 1      |             |             |             |             |             | Jobs Running: 0   | Jobs Queued: 0      |             |             |             |             |             |           |
| 🔄 Verzuim controleren | 93%               |             |             |             |             |           | Geen jobs           |                   |             |             |             |             |           | Geen jobs           |                   |             |             |             |             |           | Geen jobs           |                   |             |             |             |             | Geen jobs          |                     |             |             |             |             |             | Geen jobs         |                     |             |             |             |             |             |           |

|                     |                   |             |             |             |             |           |                            |                   |             |             |             |             |           |                     |                   |             |             |             |             |           |                     |                   |             |             |             |             |                 |                     |             |             |             |             |             |                   |                     |             |             |             |             |             |           |
|---------------------|-------------------|-------------|-------------|-------------|-------------|-----------|----------------------------|-------------------|-------------|-------------|-------------|-------------|-----------|---------------------|-------------------|-------------|-------------|-------------|-------------|-----------|---------------------|-------------------|-------------|-------------|-------------|-------------|-----------------|---------------------|-------------|-------------|-------------|-------------|-------------|-------------------|---------------------|-------------|-------------|-------------|-------------|-------------|-----------|
| <a href="#">043</a> | <div></div>       | <div></div> | <div></div> | <div></div> | <div></div> | 2025.12.1 | <a href="#">044</a>        | <div></div>       | <div></div> | <div></div> | <div></div> | <div></div> | 2025.12.1 | <a href="#">047</a> | <div></div>       | <div></div> | <div></div> | <div></div> | <div></div> | 2025.12.1 | <a href="#">048</a> | <div></div>       | <div></div> | <div></div> | <div></div> | <div></div> | 2025.12.1       | <a href="#">049</a> | <div></div> | <div></div> | <div></div> | <div></div> | <div></div> | 2025.12.1         | <a href="#">050</a> | <div></div> | <div></div> | <div></div> | <div></div> | <div></div> | 2025.12.1 |
| Summa               |                   |             |             |             |             |           | Techniek College Rotterdam |                   |             |             |             |             |           | Zadkine             |                   |             |             |             |             |           | Technicom           |                   |             |             |             |             | ROC van Twente  |                     |             |             |             |             |             | OPPStap           |                     |             |             |             |             |             |           |
| Piek sessies: 554   | Live sessies: 9   |             |             |             |             |           | Piek sessies: 272          | Live sessies: 13  |             |             |             |             |           | Piek sessies: 607   | Live sessies: 33  |             |             |             |             |           | Piek sessies: 28    | Live sessies: 0   |             |             |             |             | Piek sessies: 6 | Live sessies: 0     |             |             |             |             |             | Piek sessies: 160 | Live sessies: 0     |             |             |             |             |             |           |
| Duur: 0.133s        | Load average: 0.7 |             |             |             |             |           | Duur: 0.022s               | Load average: 0.1 |             |             |             |             |           | Duur: 0.091s        | Load average: 0.0 |             |             |             |             |           | Duur: 0.004s        | Load average: 0.1 |             |             |             |             | Duur: 0.002s    | Load average: 0.0   |             |             |             |             |             | Duur: 0.003s      | Load average: 0.0   |             |             |             |             |             |           |
| Jobs Running: 1     | Jobs Queued: 0    |             |             |             |             |           | Jobs Running: 0            | Jobs Queued: 0    |             |             |             |             |           | Jobs Running: 0     | Jobs Queued: 0    |             |             |             |             |           | Jobs Running: 0     | Jobs Queued: 0    |             |             |             |             | Jobs Running: 0 | Jobs Queued: 0      |             |             |             |             |             | Jobs Running: 0   | Jobs Queued: 0      |             |             |             |             |             |           |
| 🔄 Workflowautomaat  | 9%                |             |             |             |             |           | Geen jobs                  |                   |             |             |             |             |           | Geen jobs           |                   |             |             |             |             |           | Geen jobs           |                   |             |             |             |             | Geen jobs       |                     |             |             |             |             |             | Geen jobs         |                     |             |             |             |             |             |           |

|                     |                   |             |             |             |             |           |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|---------------------|-------------------|-------------|-------------|-------------|-------------|-----------|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|
| <a href="#">051</a> | <div></div>       | <div></div> | <div></div> | <div></div> | <div></div> | 2025.12.1 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| SG Bonaire          |                   |             |             |             |             |           |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| Piek sessies: 6     | Live sessies: 0   |             |             |             |             |           |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| Duur: 0.003s        | Load average: 0.1 |             |             |             |             |           |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| Jobs Running: 0     | Jobs Queued: 0    |             |             |             |             |           |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| Geen jobs           |                   |             |             |             |             |           |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |



# REQUEST TIJDEN IDENTITY

## JAVA APPLICATION

JAVA 21  
19 CLASSES  
1339 LINES  
NO DATABASE

## HTMX + BOOTSTRAP

HATEOAS  
HTML OVER THE WIRE

## QUARKUS

CLOUD NATIVE  
CONFIG  
REST + QUTE  
METRICS & HEALTH

*who was?*



```
private void crawlStatuses(Config config, ConcurrentLinkedDeque<CollectedMetrics> localMetrics)
{
    config.getEnvironments().parallelStream().forEach(environment -> {
        environment.tenants().parallelStream().forEach(tenant -> {
            var crawlResult = crawl(environment, tenant, tenant.urlStatus());
            var metrics = crawlResult.httpStatus() == 200
                ? convertCrawledStatusPageToMetrics(environment, tenant, crawlResult)
                : convertCrawledStatusPageToFailure(environment, tenant, crawlResult);

            localMetrics.add(metrics);
        });
    });
}
```

*A thread executor would  
limit the number of threads  
but now most retrievals complete  
in ~5 seconds*

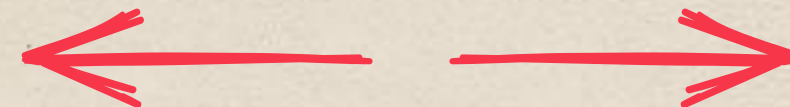
**~150 THREADS**



# What is Out of Memory?

"When we set the upper  
limit of PC-DOS at 640K, we  
thought nobody would ever  
need that much memory."

— William Gates





# OUT OF MEMORY ERRORS

## JAVA

No more Java objects can be created because of there is not enough memory available

XXXXX

## KUBERNETES

The proces in a container has asked for more memory than its resource limit

where to start?



# JAVA MEMORY AREAS



## HEAP

Where all Java objects get created, live and die.



## NON-HEAP

Meta data, compiler data, direct memory buffers, administrative memory

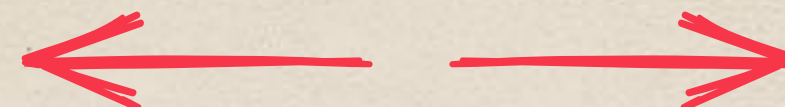


## STACK

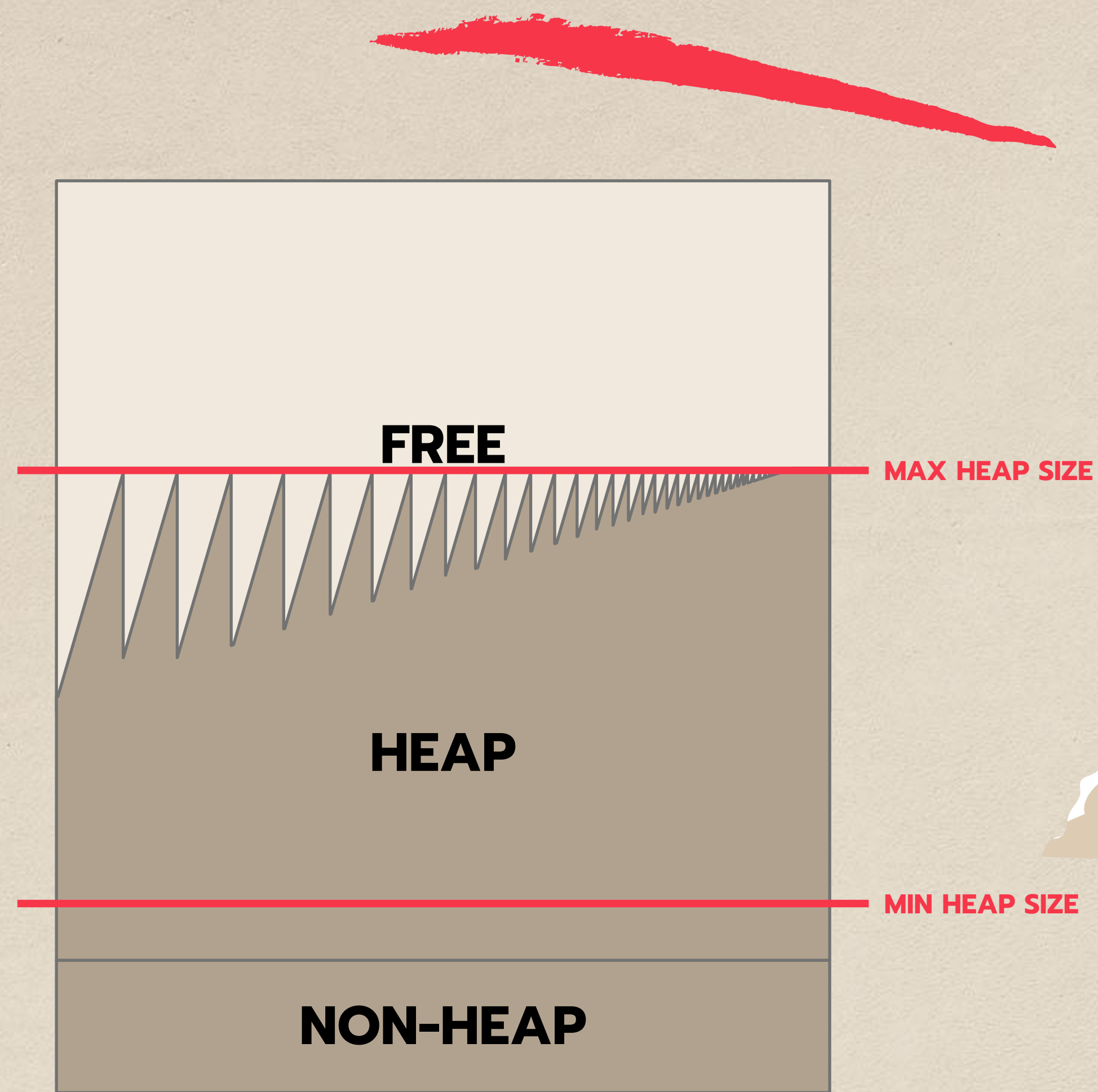
Allocations for method calls, parameters, local variables, return values for each thread



*Inspirational Quote that  
pertains to memories lost*







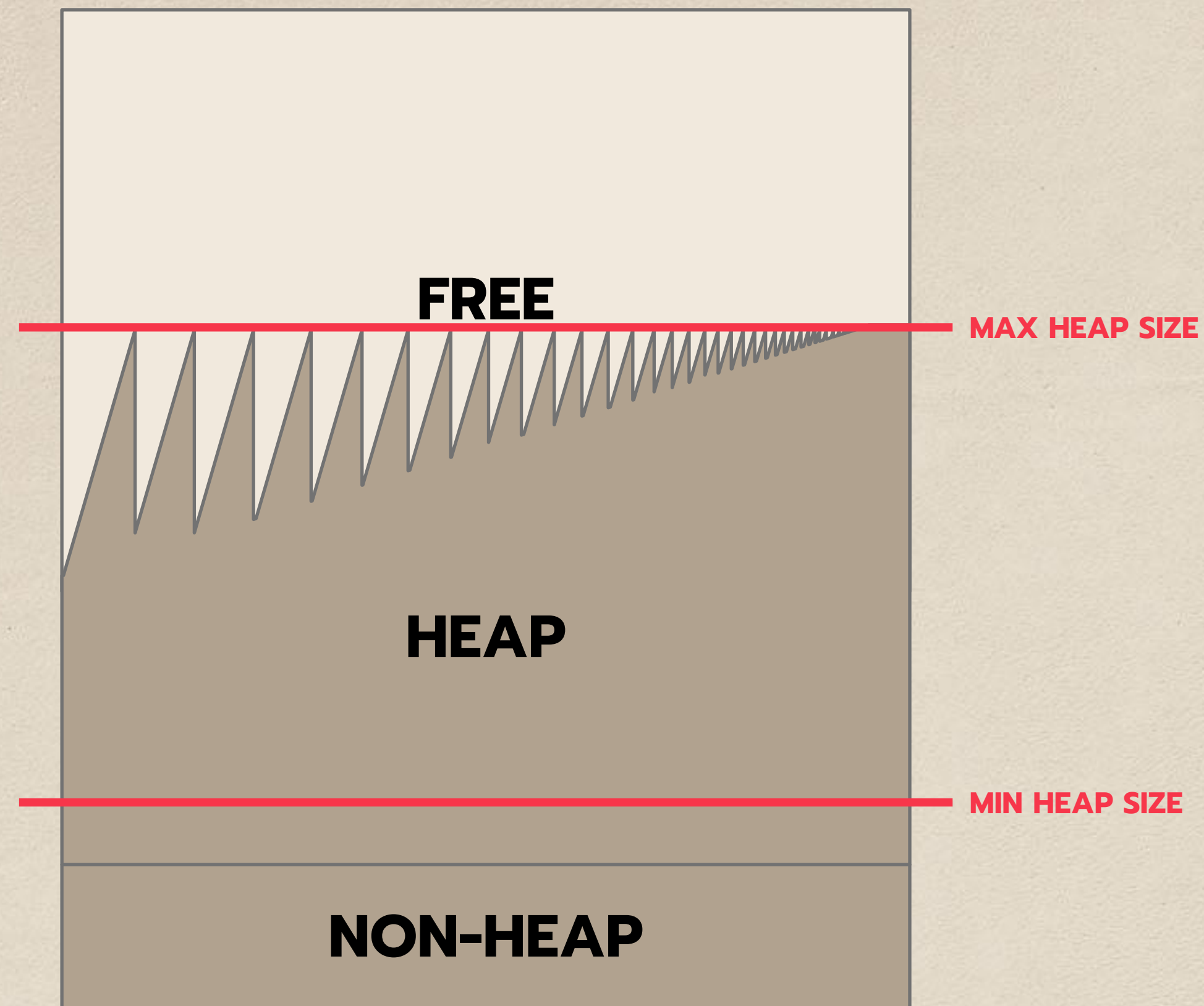
XXXX

**THE HEAP**

**JVM**







**JVM**

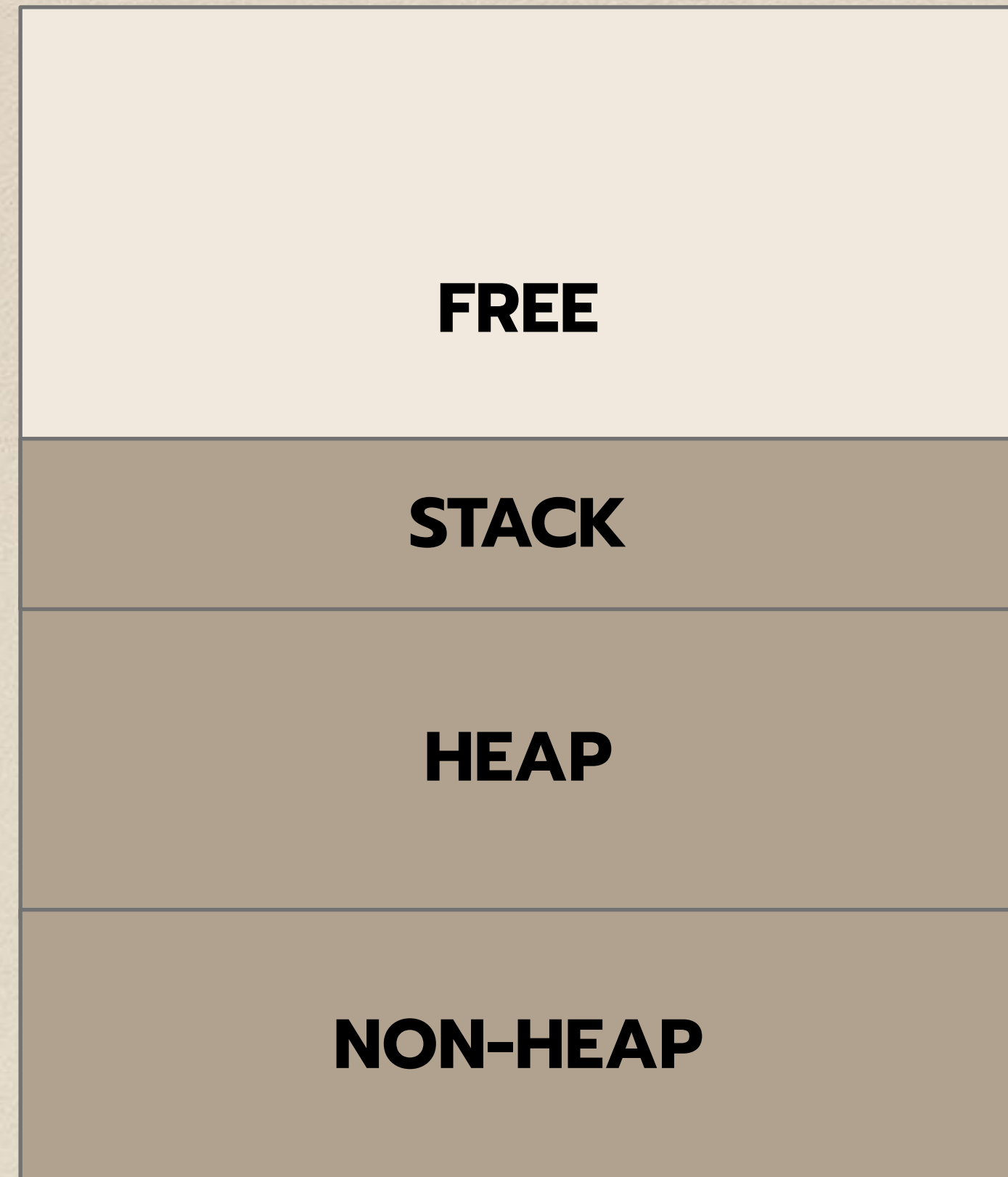
XXXX

Heap area size is determined by

- Defaults
  - Min: 1.5%
  - Max: 25%
- Absolute values
  - -Xms (min)
  - -Xmx (max)
- Relative values
  - -XX:InitialRAMPercentage
  - -XX:MaxRAMPercentage



← RESIDENT SET SIZE →



**JVM**

Non-heap area is used for:

- Metaspace (Class metadata)
- Code cache (JIT compiled)
- Direct memory buffers
- Garbage Collector data
- Stack frames

Not directly sizable

- metaspace
- direct memory buffers
- stack



**SNIPPED DUE TO FILE SIZE CONSTRAINTS**

XXXX

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NATIVE MEMORY SUMMARY

Native Memory Tracking:

(Omitting categories weighting less than 1MB)

Total: reserved=18157MB, committed=5062MB
malloc: 22MB #17353
mmap: reserved=18135MB, committed=5040MB

```
- Java Heap (reserved=16384MB, committed=4840MB)
  (mmap: reserved=16384MB, committed=4840MB, at peak)

- Class (reserved=1024MB, committed=0MB)
  (classes #534)
  ( instance classes #445, array classes #89)
  (mmap: reserved=1024MB, committed=0MB, at peak)
  ( Metadata: )
  ( reserved=64MB, committed=0MB)
  ( used=0MB)
  ( waste=0MB =60.99%)
  ( Class space:)
  ( reserved=1024MB, committed=0MB)
  ( used=0MB)
  ( waste=0MB =97.11%)

- Thread (reserved=52MB, committed=52MB)
  (threads #26)
  (stack: reserved=52MB, committed=52MB, peak=52MB)

- Code (reserved=242MB, committed=7MB)
  (mmap: reserved=242MB, committed=7MB, at peak)

- GC (reserved=372MB, committed=147MB)
  (malloc=20MB #4101) (at peak)
  (mmap: reserved=353MB, committed=127MB, at peak)

- Compiler (reserved=0MB, committed=0MB)
  (arena=0MB #4) (peak=1MB #13)

- Symbol (reserved=1MB, committed=1MB)
  (malloc=1MB #28) (at peak)

- Shared class space (reserved=16MB, committed=13MB, readonly=0MB)
  (mmap: reserved=16MB, committed=13MB, peak=13MB)

- Arena Chunk (reserved=0MB, committed=0MB)
  (malloc=0MB #59) (peak=1MB #76)

- Metaspace (reserved=64MB, committed=0MB)
  (mmap: reserved=64MB, committed=0MB, at peak)
```

```
public class Demo {
    public static void main(String[] args) {
        while (true);
    }
}
```

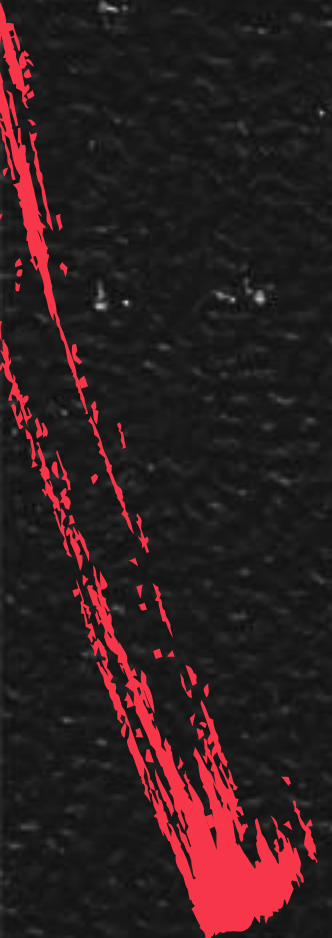
```
java -XX:NativeMemoryTracking=summary scale=MB
```

```
Java Heap (reserved=16384MB, committed=4840MB)
(mmap: reserved=16384MB, committed=4840MB, at peak)
```

```
Class (reserved=1024MB, committed=0MB)
(classes #534)
( instance classes #445, array classes #89)
(mmap: reserved=1024MB, committed=0MB, at peak)
```

```
Thread (reserved=52MB, committed=52MB)
(threads #26)
(stack: reserved=52MB, committed=52MB, peak=52MB)
```

```
Native Memory Tracking (reserved=356KB, committed=356KB)
(malloc=84KB #1521) (at peak)
(tracking overhead=271KB)
```

CREATING A KILLER PROFILE



To catch a
killer you
need to know
the killer



who was?



KUBERNETES

RESOURCE LIMITS

Requests: minimum necessary

Limits: maximum allowed

```
containers:  
- image: eduarde-requestttijden...  
  name: requestttijden  
  imagePullPolicy: Always  
  resources:  
    requests:  
      memory: 256Mi  
    limits:  
      memory: 512Mi
```

why she did it?



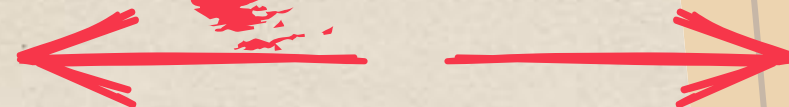
PHYSICAL MAX

RAM

FREE

BIOS

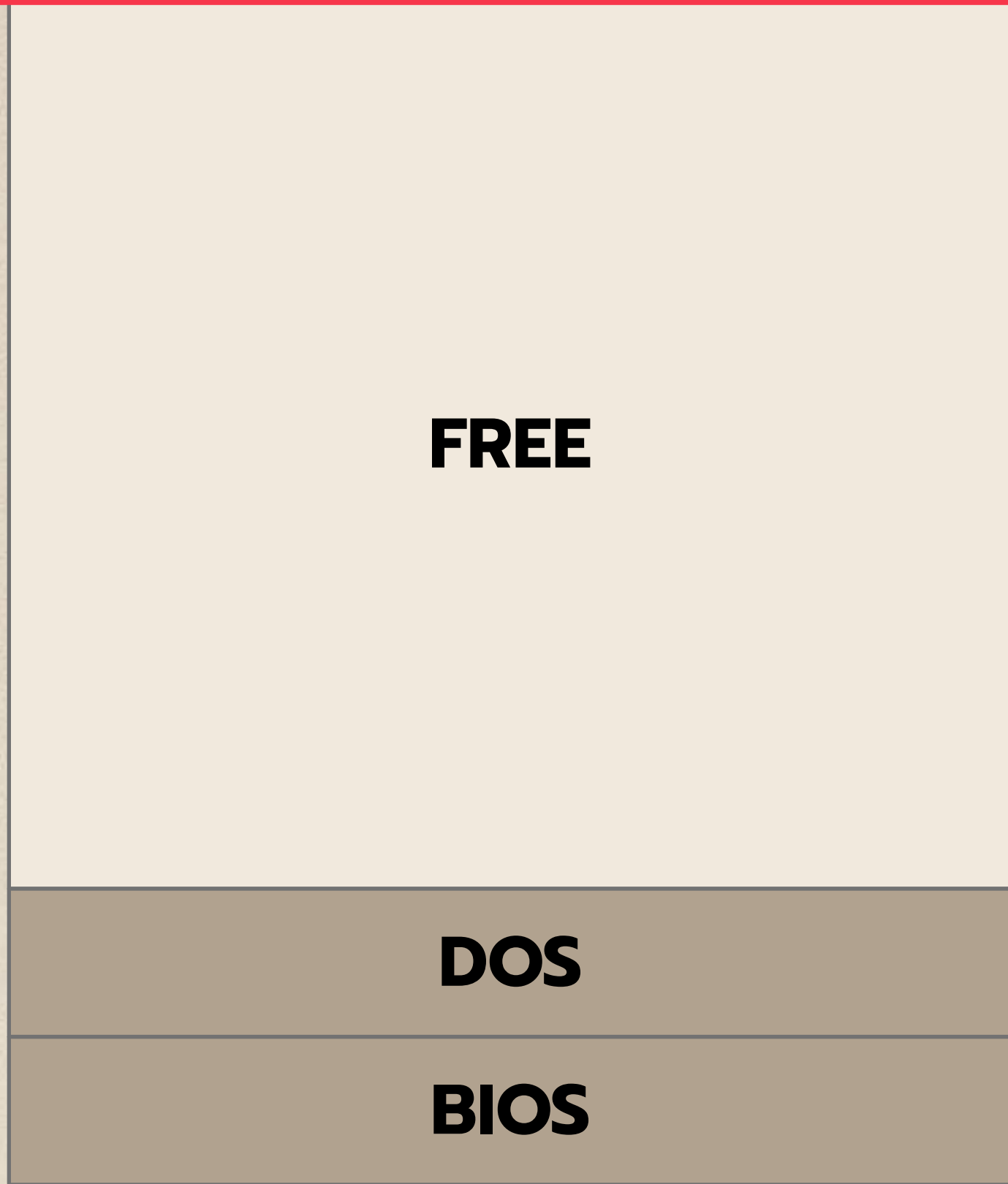
PC





PHYSICAL MAX

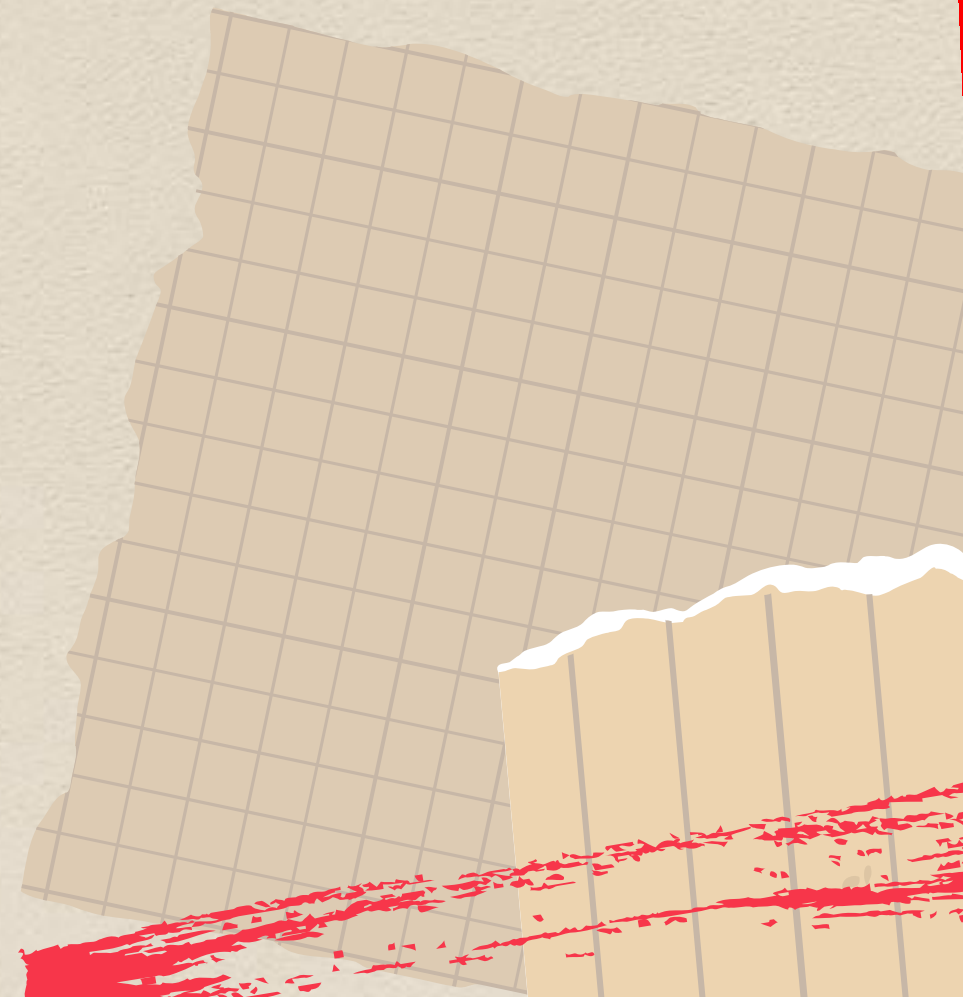
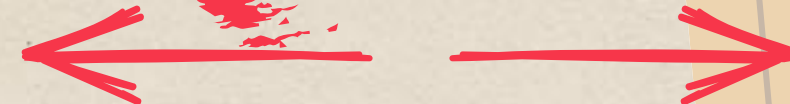
RAM



DOS

BIOS

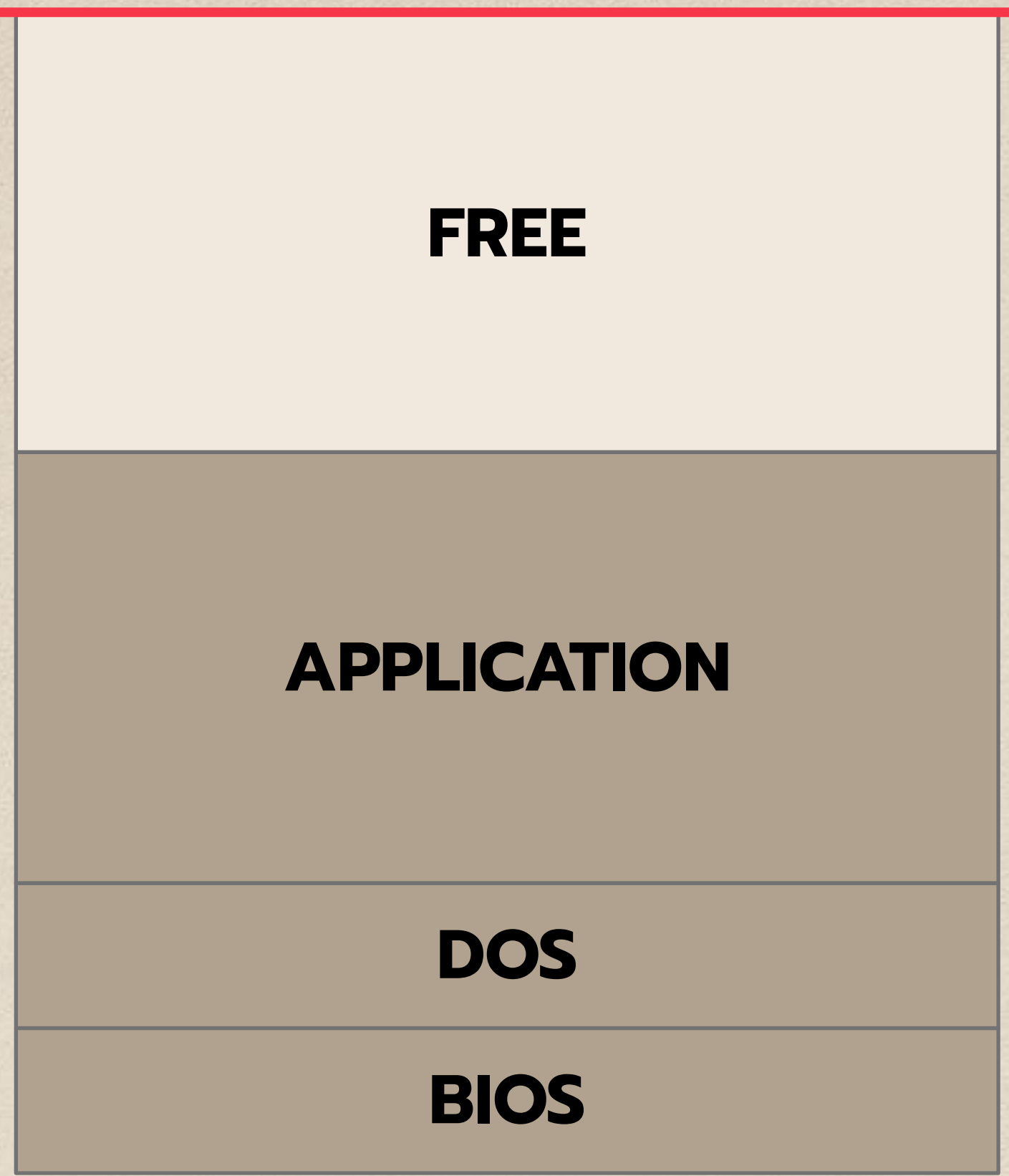
PC



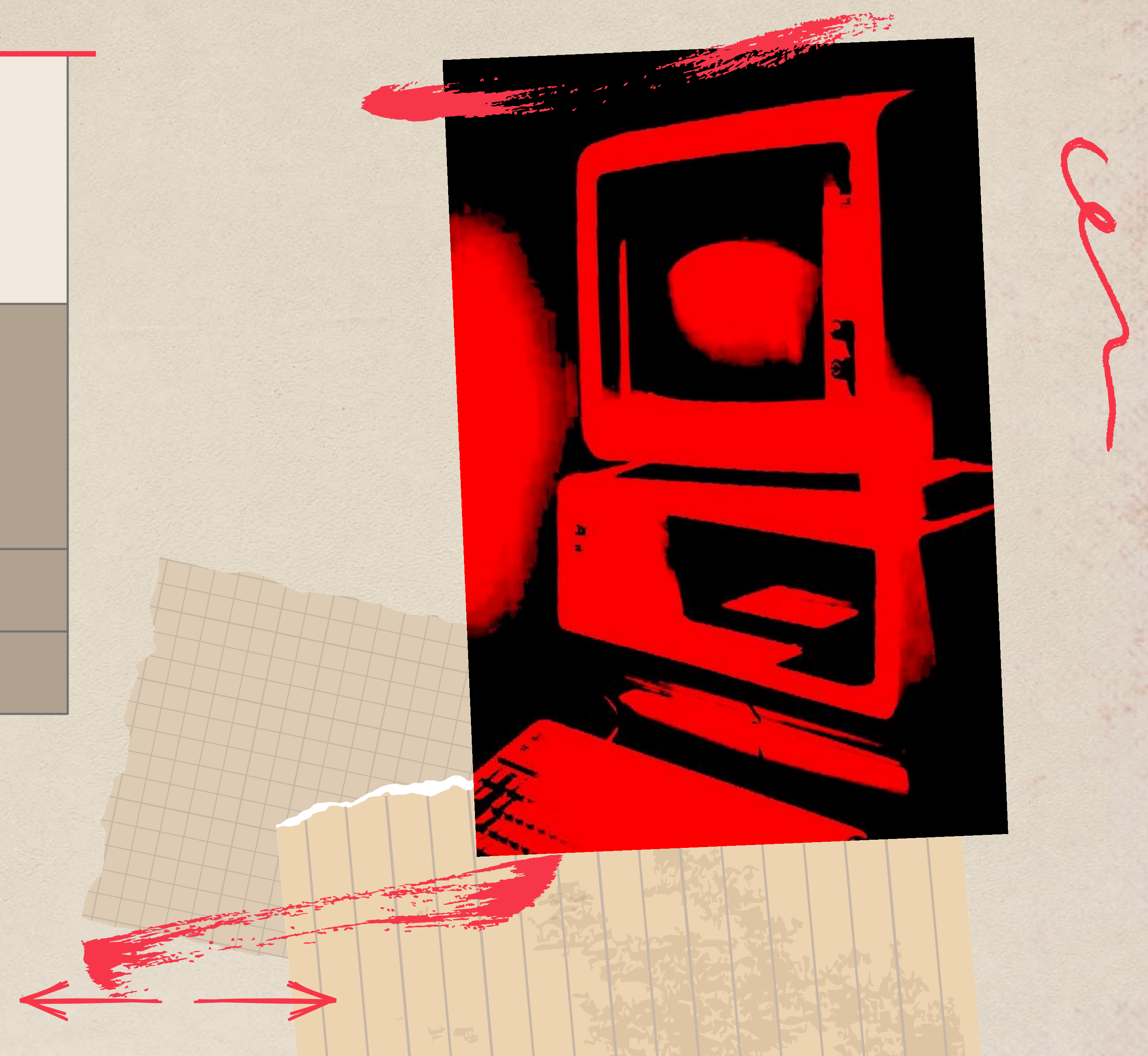
XXXX

PHYSICAL MAX

RAM



PC



XXXX

PHYSICAL MAX

RAM

FREE

APPLICATION

DOS

BIOS

PC



SNIPPED DUE TO FILE SIZE CONSTRAINTS

XXXX

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# SURVIVING THE 137-KILLER



About living that  
long container  
life





# WHAT HAVE WE LEARNED SO FAR

## CRIME EVIDENCE

Container 137 OOMKilled

512MiB memory limit

Application default settings

150 Threads

Native memory report

Grafana metrics

Application logs

Heap dumps

## JVM SETTINGS

Tune the settings to fit

## SUSPECTS

## CONTAINER LIMITS

Give more memory



# KNOW WHAT YOUR APPLICATION NEEDS

## NATIVE MEMORY SUMMARY

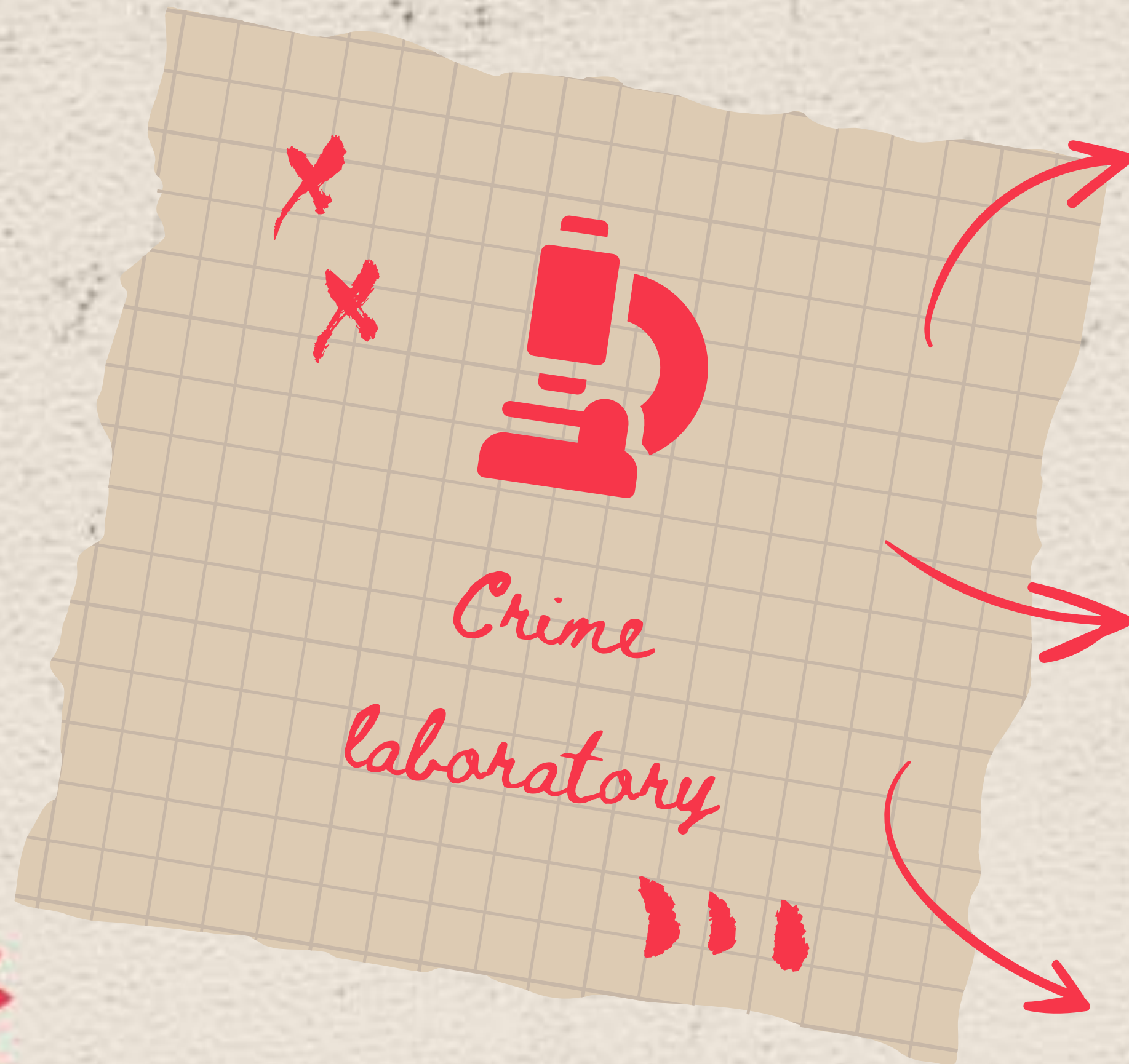
Take and analyse (Copilot/Claude/...),  
given the container restrictions

## HEAP ANALYSIS

Class Histogram, analyse a heap dump,  
tweak eden vs old gen

## NON-HEAP ANALYSIS

Number of classes, direct memory  
buffers, number of threads(!)





**SNIPPED DUE TO FILE SIZE CONSTRAINTS**

XXXX

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CATCHING THE 137-KILLER

OBSERVE

Use all tools to monitor and observe your application

ANALYSE

What type of memory causes the OOMKill?

VICTIM

Know your application's memory needs

BE VIGILENT

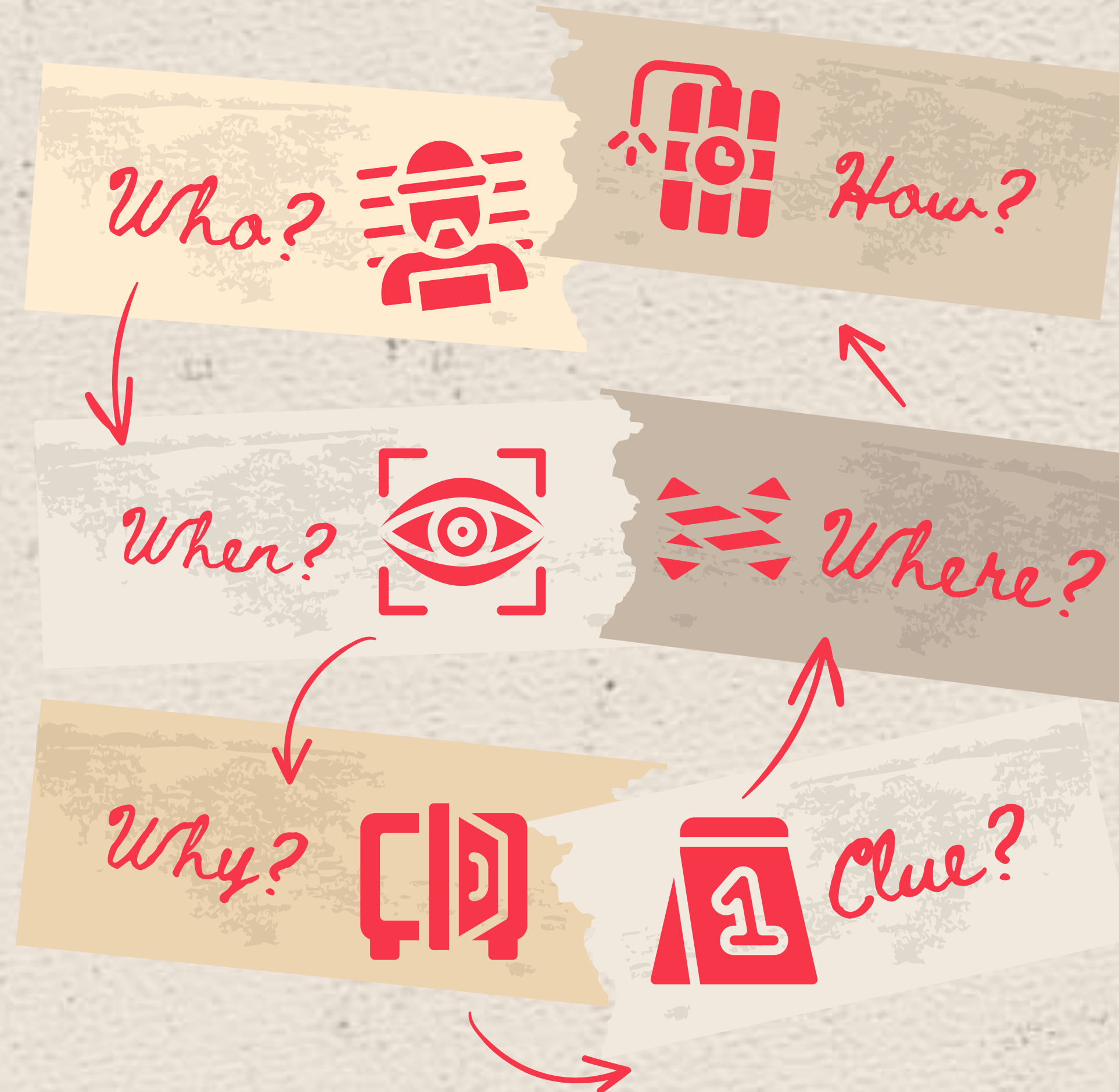
New versions will have different profiles...

ITERATE

Try the settings, monitor and adapt

PROFILE

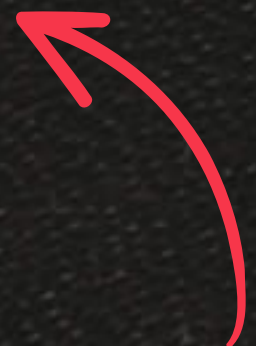
Know the settings, make sure your application works





THANKS

Did you look
closer?



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who was?

